

DEDICATED PRELIM PROGRAM(DPP)

UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022

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TEST BOOKLET SERIES

Test Code:

DPP

**TEST BOOKLET
GENERAL STUDIES
TEST -10**

E10

Time Allowed: Two Hours

Maximum Marks : 200

INSTRUCTIONS

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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TOPICS COVERED: Natural Vegetation, Soils, Agriculture, Mineral resources Energy resources Distribution of Industries Transportation, Earthquakes, Volcano, Landforms in India and across the world, Miscellaneous topics



Q.1) With reference to the 'India State of Forest Report (ISFR), 2019,' consider the following Statements:

1. In terms of absolute area, Madhya Pradesh has the largest forest cover among the states.
2. In comparison to the last assessment in 2017, there has been a decrease in forest cover in all states of North East India.
3. No Subtropical Dry Evergreen forests have been recorded in India in the latest survey report.

Which of the above statements is/are correct?

- a) 1 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1,2 and 3

Q.2) Which of the following forest types has the highest carbon stock per hectare in India?

- a) Tropical Dry Deciduous Forests
- b) Himalayan Dry Temperate Forests
- c) Tropical Thorn Forests
- d) Tropical Wet Evergreen Forests

Q.3) Consider the following statements:

1. Some of the states in India have both Tropical Evergreen and Tropical Dry Deciduous forests.
2. Some of the states in India have both Tropical Evergreen and Mangrove Forests.
3. Some of the states in India have both Tropical Evergreen forests and Alpine Vegetation.
4. Every State in India has at least some geographical area under Scrub Forests.

Which of the above statements is/are correct?

- a) 1 and 2 only
- b) 3 and 4 only
- c) 1,2 and 3 only
- d) 1,2,3 and 4

Q.4) "If you travel to a certain part of the country during the South Western Monsoons, you see a soil that is sticky and argillaceous. When you travel to the same part again in the hot dry season, the same soil has visibly deep cracks, 10-15 cm wide."

If you inquire about the nature of the soil, the farmers are likely to inform you that the soil is

- a) Alluvial Soil
- b) Black Soil
- c) Red Soil
- d) Peaty Soil

Q.5) In the context of forest surveys, 'Margalef index', and the 'The Shannon Wiener Index' are used for measuring the

- a) Total biomass of a forest
- b) Carbon footprint of a forest
- c) Albedo of a forest
- d) Species diversity in a forest

Q.6) Consider the following statements regarding laterite soils of India :

1. Laterite soils are generally red in colour
2. Laterite soils are poor in nitrogen and potash
3. Laterite soils are well developed in Rajasthan and UP
4. They are excellent for construction purposes

Which of the statements given above is /are correct?

- a) 1 only
- b) 2,3 and 4
- c) 1 and 4 only
- d) 1,2 and 4

Q.7) With reference to Indian Soils, which of the following statements is not correct ?

- a) Of all the soils, alluvial soils occupy the highest percentage of total geographic area of the country.
- b) Peaty soils act as naturally good carbon sinks in the arid regions of India.
- c) Excess and unscientific irrigation can lead to salinization of soil
- d) Forest and mountain soils have naturally high humus content but are deficient in Lime content.

Q.8) Which of the following has the highest Bamboo bearing area in the country?

- a) Odisha
- b) Arunachal Pradesh
- c) Madhya Pradesh
- d) Chattisgarh

Q.9) In the context of Indian agriculture, how can Drones be used for increasing efficiency and productivity levels?

1. Thermal, multispectral and hyperspectral Images from drones can locate underperforming areas in a farm and enable better crop management.
2. Remotely accessing fields via drones can help detect bacterial or fungal infections early.



3. Drones can be equipped with tools that give them the ability to spray the precise amount of chemicals from a precise altitude.
4. Drones can be used for agriculture mapping and planning.

Choose the correct Ans: using the code given below:

- a) 1 and 2 only
- b) 1,2 and 3 only
- c) 3 and 4 only
- d) 1,2,3 and 4

Q.10) Which of the following most appropriately describes the phenomenon of "Pesticide Treadmill"?

- a) Biomagnification of pesticides in the tissues of tolerant organisms at successively higher levels in a food chain.
- b) Resistance in plant species due to erratic and prolonged use of pesticide.
- c) Increase in doses of pesticides to prevent the resurgence of pests that were being controlled with smaller doses earlier.
- d) The sale or use of an adulterated, banned or misbranded pesticide.

Q.11) With reference to agriculture and livestock rearing in India, how can the principles of "Precision farming", often seen in the news, be used in the immediate pesticide. New future?

1. Driverless machinery can be programmed by GPS to spread fertilizer or plough land.
2. With the internet of things, it is possible to monitor a honey bee colony's health via wireless sensors.
3. Machine learning can be employed to process information and data recorded by drones and robots and send appropriate actions back to these devices.
4. Telematics can be employed to remotely control autonomous machines.

Choose the correct Ans: using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1,3 and 4 only
- d) 1,2,3 and 4

Q.12) Which of the following most appropriately describe the terms 'Harrowing' and 'Planking', sometimes seen in reference to agriculture practices in India ?

- a) These are faulty practices carried out in primitive agrarian societies that negatively impact the fertilizer absorption capacity of Soils.
- b) They are practices carried out to smoothen the hard clods in soil and uproot weeds and stubbles.
- c) They are techniques specifically designed for Direct Seeding of Rice only.
- d) These are methods of propagation in sugarcane, similar to the process of ratooning.

Q.13) Which of the following describes the phenomenon of 'Terminator Technology', sometimes seen in the context of Transgenic crops?

- a) A gene editing technology that completely eliminates mutation of disease-causing viruses in plants and helps overcome challenges in breeding.
- b) Genetic modification of crops to make them produce sterile seeds in the second generation.
- c) Genetic engineering of crops to produce an enzyme that enables them to remain healthy even after being sprayed with glyphosate.
- d) None of the above

Q.14) Consider the following statements:

1. The area under irrigation for rice cultivation in India has steadily increased in the last decade.
2. The absolute area under wheat cultivation in India has steadily increased in the last decade.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the above

Q.15) Consider the following:

1. Root & tuber crops
2. Coconut
3. Cocoa
4. Coffee
5. Bamboo
6. Mushrooms

Which of the above are included under the 'Mission for Integrated Development of Horticulture (MIDH)'?

- a) 1,2 and 6 only
- b) 1,3,4 and 5 only
- c) 2,3,4 and 6 only
- d) 1,2,3,5 and 6 only

Q.16) "The crop is tropical in nature but can be grown in subtropical areas as well. It requires well distributed rainfall of 50-100 centimeters for its



growth. Curing is an important process that makes the crop ready for the market. A well drained fertile sandy loam is suitable for its growth. “

Which one of the following is the crop?

- a) Tobacco
- b) Tea
- c) Maize
- d) Mustard

Q.17) Which of the following are Rabi Crops?

- 1. Rapeseed
- 2. Linseed
- 3. Lentil
- 4. Barley

Choose the correct code using the code given below

- a) 1 and 2 only
- b) 3 and 4 only
- c) 1,2 and 3 only
- d) 1,2,3 and 4

Q.18) A state in India has the following characteristics:

- 1. It is the largest producer of nutri cereals in India.
- 2. The climate ranges from arid to semi-arid due to high variability in South Western Monsoon rainfall.
- 3. The agrarian community relies on traditional water harvesting structures such as *Jhalras* and *Khadeen*.

Which of the following states has all the features mentioned above?

- a) Madhya Pradesh
- b) Rajasthan
- c) Haryana
- d) Tamil Nadu

Q.19) With reference to direct seeding of rice (DSR), consider the following statements:

- 1. Direct-seeded rice matures 7 to 10 days earlier than transplanted rice.
- 2. There is greater crop-weed competition in DSR than that in transplanted rice.
- 3. The seed requirement for DSR is lower than that required in the transplantation method.

Which of the above statements is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3

Q.20) Which of the following made up the highest value among agricultural exports from India in the financial year 2020-21?

- a) Basmati Rice
- b) Castor Oil
- c) Cotton
- d) Tea

Q.21) With reference to ‘Agri-business farming’, Consider the following statements

- 1. It is a type of commercial farming managed and financed by a group of small and medium farmers who pool their lands and resources.
- 2. It is highly mechanized and relies on chemicals and fertilizers.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2 only
- d) None of the above

Q.22) Consider the following statements:

- 1. Majority of lignite deposits in India occur in the Chotanagpur plateau.
- 2. There is no trace of anthracite coal deposits in India.
- 3. Peat deposits have been found in the Nilgiri hills of Western Ghats.
- 4. The Gondwana Coal has traces of sulphur, but is completely free of moisture.

Which of the above statements is/are correct?

- a) 3 only
- b) 1 and 3 only
- c) 3 and 4 only
- d) 1,2 and 4 only

Q.23) In which of the following regions has The Ministry of Petroleum and Natural Gas *not* found prospective shale gas reserves in India?

- a) Gangetic Basin
- b) Assam-Arakan Basin
- c) Cauvery Basin
- d) Bombay High Field

Q.24) Consider the following:

- 1. B-heavy Molasses
- 2. C-heavy Molasses
- 3. Surplus rice with Food Corporation of India (FCI)
- 4. Damaged food grain
- 5. Maize
- 6. Sugarcane juice



Which of the above categories of feed stock can be used for production of ethanol under the Ethanol Blending Programme of the Ministry of Petroleum and Natural Gas ?

- a) 1,2,4 and 6 only
- b) 1,3,4,5 and 6 only
- c) 2,3 and 5 only
- d) 1,2,3,4,5 and 6

Q.25) "This 700 MWe Nuclear power plant in Gujarat recently became the first Nuclear reactor of its kind to achieve criticality i.e. it reached the normal operating condition of a reactor and is set to generate power."

The Nuclear Power Plant, referred to in the above excerpt, is

- a) Tarapur Nuclear Reactor
- b) Rawatbhata Nuclear Reactor
- c) Narora Nuclear Reactor
- d) Kakrapar Nuclear Reactor

Q.26) Consider the following Statements:

- 1. The Per capita Energy Consumption in India has steadily increased in the last five years.
- 2. The Energy Intensity in India has steadily decreased in the last five years.
- 3. Recently, solar power has overtaken wind power in terms of total installed capacity.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3

Q.27) Consider the following raw materials used in iron and steel industries:

- 1. Iron ore
- 2. Coking coal
- 3. Limestone
- 4. Manganese
- 5. Dolomite

Which of the above raw materials are weight losing in nature ?

- a) 1 and 2 only
- b) 1,2,3 and 5 only
- c) 3,4 and 5 only
- d) 1,2,3,4 and 5

Q.28) Which of the following industries can be categorised as Footloose industries?

- 1. Mobile and Computer manufacturing
- 2. Diamond Cutting

- 3. Aluminium Smelting
- 4. Fertilizers Industry
- 5. Automobile Manufacturing

Select the correct Ans: using the codes given below

- a) 1,2 and 5 only
- b) 1,3 and 4 only
- c) 2,3 and 4 only
- d) 1,2,3,4 and 5

Q.29) With reference to Sugar industry in India, consider the following statements:

- 1. While sugar mills are generally located close to sugarcane growing areas, sugar refineries can be set up at relatively distant locations.
- 2. In India, the pricing of sugarcane is governed by the statutory provisions of the Sugarcane (Control) Order, 1966 issued under the Essential Commodities Act, 1955.

Which of the above statements is/are incorrect ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the above

Q.30) Which of the following industries can be categorised as Sunrise industries ?

- 1. Hybrid Electric Vehicles
- 2. Green Hydrogen
- 3. Genome Sequencing
- 4. Thermal Power Plants
- 5. 5G Mobile Network
- 6. Technical Textiles

Select the correct Ans: using the code given below:

- a) 1,2,3 and 4 only
- b) 1,4,5 and 6 only
- c) 1,2,3,5 and 6 only
- d) 1,2,3,4,5 and 6

Q.31) "This pipeline runs through the states of Gujarat, Madhya Pradesh and Uttar Pradesh. It has been constructed by the Gas Authority of India Limited and is 1750 kilometres long. It connects the Bombay High with Northern India."

Which of the following pipelines in India is best described by the above excerpt ?

- a) Salaya-Koyali-Mathura Pipeline
- b) Hajira-Bijapur-Jagdipshpur Gas Pipeline
- c) Jamnagar-Loni LPG Pipeline
- d) Kandla-Bhatinda Pipeline

Q.32) Consider the following statements:



1. When completed, the Western Dedicated Freight Corridor(WDFC) will traverse through a total of five states, including the destination and origin.
2. When completed, the Eastern Dedicated Freight Corridor (EDFC) will traverse through a total of four states, excluding the destination and origin.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the Above

Q.33) To promote inland water transport in the country as an economical, environment friendly supplementary mode of transport to rail and road, 111 inland waterways have been declared as 'National Waterways' in India. In this context consider the following statements:

1. The declaration of National waterways was made via an executive order.
2. National Waterway-4 traverses through more states than National Waterway-1.
3. Recently in January 2021, all waterways have become fully operational.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) 1,2 and 3
- d) None of the Above

Q.34) Consider the following statements:

1. The railways' share in transportation of surface freight in India has declined since independence.
2. Rail transportation is more energy efficient than road transport.
3. In India, RORO services are provided via railways only.

Which of the above statements is/are correct?

- a) 2 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 1,2 and 3

Q.35) Which of the following statement is/are correct ?

1. Earthquake swarm is a series of high magnitude earthquakes occurring in a localised region over a period of time ranging from days to weeks to even month
2. Earthquake swarm can serve as marker for location of flowing magma

3. Surface waves which are responsible for the most destruction are high frequency, short wavelength transverse waves

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) 1 & 3 only
- d) 1, 2 & 3 only

Q.36) With reference to the measuring of Earthquakes, consider the following statements :

1. Richter scale or the Magnitude Scale measures the Magnitude of the damage caused by the earthquake
2. Mercalli Scale or the Intensity Scale measures the Intensity of the energy released during the Earthquake
3. Richter scale has a variation of 1 - 10 while Mercalli scale has of 1-12

Which of the following statement is/are correct?

- a) 1 & 2 only
- b) 3 only
- c) 1, 2 & 3
- d) None of the above

Q.37) Which of the following statements are Correct regarding Earthquake Prone Zones in India

- a) Ministry of Earth Science has divided India into four Seismic Zone
- b) Zone V is the most seismically active zone while Zone II is the least seismically active zone
- c) Area covered under Zone III is the most while Zone V covers the least share of total seismically active area
- d) Zones are divided on the basis of Modified Richter Scale which measures the magnitude of Earthquake.

Q.38) Consider the following statement is Incorrect regarding Earthquake Prone Zone

- a) About 2/3rd of India's land area is under threat of moderate to severe earthquakes
- b) Zone V is entirely confined to Himalayan Belt and North Eastern States
- c) Lakshadweep comes entirely under Zone III while Southern states of Tamilnadu, Kerala and Telangana shares area under both Zone II and Zone III
- d) Zone IV is mostly confined to Himalayan Belt and parts of its foothills and plain area with some small pockets along both Eastern and Western coasts



Q.39) With Reference to Tsunami, consider the following statement

1. Tsunamis are a series of very long wavelength and period created in ocean by impulsive disturbance.
2. India volunteered to Join International Tsunami Warning system after December 2004 Tsunami disaster
3. 2004 Tsunami disaster involved the tectonic event where Indian plate went under Burma plate

Which of the following statement is/are correct?

- a) 1 only
- b) 1 & 2 only
- c) 1 & 3 only
- d) 1,2 & 3

Q.40) Consider the following statement about Tsunamis

1. Tsunami waves travel at different speeds in water - faster in deep water & slower in shallow water
2. Amplitude of tsunami waves is more in deep water than in shallow water

Choose the correct statement/s

- a) 1 only
- b) 2 only
- c) Both 1 & 2
- d) Neither 1 nor 2

Q.41) With reference to volcanic Landforms. , Consider the following statements

1. Sills, Dykes and Batholith are intrusive Volcanic landform of Solidified magma
2. Country Rocks gets metamorphosed by heat from Phacolith
3. Batholith is a part of original magma chamber

Which of the following Statement is/are true?

- a) 1 only
- b) 2 only
- c) 2 & 3 only
- d) 1, 2 & 3

Q.42) Consider the following statements

1. Hot springs, Geysers & Fumaroles display release of underground energy
2. Both Geysers and Fumaroles are characterized by release of steam
3. Hot springs Feature spring heated by geothermally heated groundwater while Geysers are spring featured by turbulent ejection of water
4. Geysers show distinct colour due to presence of cyanobacteria while Hot springs are colourful due to presence of silica deposit

Which of the following statement is/are correct?

- a) 1,2 & 3 only
- b) 2, 3 & 4 only
- c) 1, 3 & 4 only
- d) 1, 2 & 4 only

Q.43) With reference to Hotspot Volcanism, consider the following statements :

1. Hotspots are relatively fixed in comparison to the Tectonic plates
2. It is different from other type of Volcanism as it occurs not at margins but at the interior parts of the Lithospheric plates
3. Term "Hotspot" in Hotspot Volcanism refers to some mostly fixed spot(area) in Earth's Mantle where hot plumes rises upwards through convection

Choose the correct statement

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 only
- d) 1, 2 & 3

Q.44) Which of the following statements is Correct ?

- a) Hotspot Volcanism forms a chain due to the difference that the hotspot are comparatively fixed while the plate moves
- b) Hawaiian eruption are characterized by violent eruptions while Vesuvius Volcanism is quiet
- c) Hotspots can't develop beneath continents
- d) Mantle plumes are exceptionally hot areas fixed at shallow depth from crust

Q.45) With reference to Earthquake and Volcanic Eruption , which of the following statement is Incorrect

- a) There has been no active volcanic activity in India after 1995
- b) Mediterranean region is second most active volcanic activity zone next to Circum Pacific belt
- c) Ring of Fire or Circum Pacific belt is a subduction zone in entirety along its stretch
- d) Recent research shows cooling off of Pacific plate in ring of fire

Q.46) Consider the following statement

1. Gravity as a driving force is common in all denudational processes
2. Denudational processes are always an in-situ process

Which of the following statement is/are correct?

- a) 1 only
- b) 2 only



- c) Both 1 & 2
- d) Neither 1 nor 2

Q.47) Consider the following statements

1. Exfoliation domes are large and pointed formed as a result of weathering
2. Drastic diurnal temperature change aid the process of exfoliation making Dry climate and high elevation most effective location for exfoliation
3. Shattering produces very round pieces of rock with blunt edges

Which of the following statements is/are correct

- a) 1 only
- b) 2 & 3 only
- c) 2 only
- d) 1 & 3 only

Q.48) With reference to denudational process, consider the following statements

1. Weathering aids both in Soil formation and Enrichment
2. Weathering is a prerequisite for mass movement
3. Mass movement doesn't involve any geomorphic agents

Choose the correct Statements

- a) 1 only
- b) 2 & 3 only
- c) 1 & 3 only
- d) None of Above

Q.49) With reference to Mass movement, consider the following statements

1. Talus are common at the base of Rock or Debris slide
2. Slide is different from Slump as there is no rotation of sliding rock mass along a curved surface

Which of the following statement is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.50) With reference to Glacier Landforms, consider the following statements

1. Glaciers form steep valleys with straight sides and flat bottom
2. Drumlins are aggradational Landforms with smooth surface at both side while Roche Moutonne are degradational landforms with smooth surface only at upstream side

3. Composition of a Glacial Landforms may not match with the characteristics feature of the underlying region

Which of the above are correct?

- a) 1 & 2 only
- b) 1 & 3 only
- c) 2 only
- d) 1, 2 & 3 only

Q.51) With reference to Riverine landform, consider the following statements

1. Gorges are almost equal in width at both top and bottom unlike Canyons which is wider at top than its bottom
2. Retreat of waterfall can result into formation of a gorge with steep sides
3. Incised meanders can be both symmetric or asymmetric. Entrenched meanders are symmetrical while Ingrown members are symmetrical
4. Alluvial Fans doesn't have clear stratification while Deltas have clear Stratification
5. Point bars or meander bars are sediments deposited in semicircular fashion by flowing rivers
6. Middle course of river is characterised by less friction in water flow in the river channel

Which of the above are correct?

- a) 1, 3, 5 & 6
- b) 2, 4, 5 & 6
- c) 1, 2, 3, 4 & 6
- d) 4, 5 & 6

Q.52) With Reference to Karst landform and Marine landform, consider the following statements

1. Karst landform constitute about 8% of Earth's land surface and is limited to area where comparatively soluble rocks exist
2. Disappearance of streams in limestone regions are related to sinkhole, swallow hole and blind valley
3. Submerged coasts are dominated by depositional feature while Emerged coast are dominated by Erosional features
4. Tombolo is a depositional marine landform in which an Island is attached to mainland by a narrow piece of land

Which of the following statements is/are correct?

- a) 1, 2 & 4
- b) 2, 3 & 4
- c) 1, 2 & 3
- d) 1, 2, 3 & 4



Q.53) Consider the following pairs:

Soils	Deficient in
1. Red soils	- Nitrogen
2. Laterite soils	- Lime
3. Black soils	- phosphates

Which of the above pairs is/are correctly matched?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1,2 and 3

Q.54) which of the following statements are correct about Saline-Alkaline soils?

- 1. The top soil is deficient in salt particles.
- 2. They are common in areas irrigated with canal and high subsoil water tables.
- 3. They occur in nature because of human activities and even naturally.

Select the correct Ans: using the codes given below:

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1,2 and 3

Q.55) Which of the following are the characteristics of coniferous forests?

- 1. The leaves are broad leafed and trees have a large spread which provides shade during summer months.
- 2. There is no annual replacement of leaves.
- 3. They can be found in both tropical and temperate regions.

Select the correct Ans: using the code given below:

- a) 1, 2 and 3
- b) 2 and 3 only
- c) 1 and 2 only
- d) 1 and 3 only

Q.56) Ber, neem, babool, date palm are common trees found in which type of vegetation?

- a) Moist deciduous forests
- b) Tropical evergreen forests
- c) Dry deciduous forests
- d) Tropical desert forests

Q.57) With reference to the "Laterite soils", which of the following statements is/are correct?

- 1. They develop in regions of high temperature, and low rainfall.

- 2. Soil is majorly devoid of Humus content.
- 3. North-western India has the major portion of the laterite soils in India

Select the correct answer using the code given below:

- a) 1, and 2 only
- b) 2 only
- c) 1, and 3 only
- d) 3 only

Q.58) Which of the following are the ideal conditions for the formation of Saline soils?

- 1. Seawater intrusion
- 2. Lack of irrigation facilities
- 3. Heavy drainage conductivity of the soil
- 4. Waterlogged areas
- 5. Dry climatic conditions

Select the correct Answer using the code given below

- a) 1, 2, and 5 only
- b) 1, 4, and 5 only
- c) 2, 3, and 4 only
- d) 1, 2, 3, 4, and 5

Q.59) For which of the following reasons, the farmers are advised to add "Gypsum" to the soil for cultivation in "Saline soils"?

- a) It increases the water-holding capacity of the soil
- b) It fills up the air gaps in the soil so that air cannot be entered into it.
- c) It leaches out the sodium chloride proportion from the soil
- d) It promotes capillary action in the soil.

Q.60) With reference to the Tropical evergreen forests, consider the following statements:

- 1. Tropical evergreen forests are dense and multi-layered.
- 2. They are found in regions receiving rainfall of 150 to 200 cm.
- 3. Single species dominate in the evergreen forests.
- 4. In India they are found extensively in Andaman and Nicobar.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 1 and 4 only
- c) 2, 3 and 4 only
- d) 1, 2, 3 and 4 only

Q.61) Consider the following pairs:

Vegetation Types

Forest Species



- | | |
|---------------------------------|----------|
| 1. Tropical Thorn Forest | Neem |
| 2. Subtropical Broadleaf forest | Chestnut |
| 3. Moist deciduous forest | Rosewood |
| 4. Tropical Evergreen Forest | Teak |

Which of the pairs given above are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1, 2 and 3 only
- d) 1, 2, 3 and 4

Q.62) Consider the following statements:

- 1. Southern slope of the Himalaya carries a thicker vegetation cover compared to the Northern slope of the Himalaya.
- 2. Western Himalaya has higher biodiversity and higher density compared to Eastern Himalaya.
- 3. Sholas are the Grassland Forest in the leeward side of Nilgiris, Anaimalai and Palani Hills.

Which of the statements given above is/are Incorrect?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.63) consider the following statements:

- 1. As per Jute Packaging Materials Act, 1987, hundred percent of food grains and twenty percent of sugar is to be mandatorily packed in diversified jute bags.
- 2. Jute thrives in tropical highland areas with high humidity.
- 3. India is the world's largest producer of raw jute.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.64) Consider the following statements about cotton production in India:

- 1. SUVIN variety of cotton is a result of hybridisation but unlike others, its seeds can be reliably saved and used again.
- 2. Cotton Technical Assistance Program is implemented by India to assist African countries in increasing their competitiveness.

Which of the statements given above is/are correct?

- a) 1 only

- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.65) Consider the following statements.

- 1. Coal bed methane (CBM) can be accessed from the same source of Offshore Natural gas.
- 2. Shale gas and oil can be extracted using a specific process of Hydraulic fracking and horizontal drilling.
- 3. Adding guar gum increases viscosity of water and makes the hydraulic fracturing process more efficient.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) 1, 2 and 3 only

Q.66) With reference to Solar energy in India, Consider the following statements.

- 1. Solar Thermal Power plants directly convert Solar thermal energy into Electrical energy.
- 2. Solar PhotoVoltaic (PV) cell converts solar energy into heat energy which is further utilised to produce electricity.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.67) Consider the following statements :

- 1. Coal is a Major mineral and Lignite is a minor mineral as per Mines and Mineral Development and Regulation Act 1957.
- 2. India's more than 90 % Coal Reserve and Production comes from Gondwana coal fields.

Which of the statements given above is/are NOT correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.68) With reference to the atomic mineral resources In India, Consider the following statements.

- 1. India has sufficient Uranium resources to meet her domestic demands of energy production.



2. India has huge Thorium resources in the monazite sands of Kerala which can be directly used in first stage nuclear reactors.
3. India is totally dependent on imports to meet her demand of Beryllium oxide, used as moderator in Nuclear Power Plants.

Which of the statements given above is/are not correct?

- a) 1 and 2 only
- b) 2 only
- c) 3 only
- d) 1, 2 and 3

Q.69) Consider the following statements:

1. World's largest concentration of lithium resources is found in the triangle of China, Japan and South Korea.
2. India is the largest Consumer as well as largest importer of Lithium resources.
3. Recently the Department of Atomic Energy found the presence of lithium resources in the igneous rocks of the Margalla-Allapatna region of Karnataka.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 only
- c) 3 only
- d) 1 and 3 only

Q.70) Consider the following statements.

1. In India, state governments notify minor minerals and the central government has no power to frame rules for that.
2. Madhya Pradesh is the only diamond producing state in India.
3. Karnataka and Andhra Pradesh are the largest gold producing states in India.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3 only

Q.71) World Heritage Sites in India is/are under the protection of which of the following organizations?

1. Ministry of Environment, Forest and Climate Changes
2. Municipal Corporations
3. Archaeological Survey of India
4. Ministry of Railways

Select the correct Ans: from the codes given below:

- a) 1, 2 and 4 only
- b) 3 only
- c) 3 and 4 only
- d) 1, 2, 3 and 4

Q.72) Consider the following statements with respect to the Returning Officer of an election in India.

1. Role of a Returning Officer during an election is decided by the Election Commission as per the requirement of an election.
2. A Returning Officer is responsible for overseeing the election in a constituency, or sometimes in two constituencies, as directed by the Election Commission (EC).
3. He/She is responsible for designating counting centres and counting of the votes during an election.

Select the correct Ans: using the codes given below.

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.73) For creation/alteration of a new district within a state, permission of which of the following is/are required?

1. Union Home Ministry.
2. Ministry of Earth Sciences
3. Intelligence Bureau
4. Department of Post
5. Geographical Survey of India
6. Railway Ministry.

Select the correct Answer from the codes given below.

- a) 1 only
- b) 2, 3 and 5 only
- c) 1, 2, 3, 4 and 6 only
- d) None of the above.

Q.74) With reference to the initiative or an alliance "17 + 1", consider the following statements:

1. It is a China led initiative that focuses on infrastructure projects such as bridges, motorways, railway lines and modernisation of ports in the member states.
2. The members consist of China and 16 European Union States only.



Which of the above given statements is/are Incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.75) Ottawa Declaration is related to the -

- a) Trade Relations between European Union and North America.
- b) Establishment of the Arctic Council
- c) Creation of the G-7 group
- d) Protection of World Cultural and Natural Heritage

Q.76) Which of the following is not a provision under the PM CARES For Children- Empowerment of COVID Affected Children launched for support & empowerment of Covid affected children?

- a) Children who lost their parents due to Covid -19 will get a monthly stipend once they turn 18 and a fund of Rs 10 lakh when they turn 23 from PM CARES FUND.
- b) Such children if admitted in a private school, the fees as per the RTE norms will be given from the PM CARES FUND.
- c) The children affected will be given a loan for higher education from the PM CARES FUND as per special provisions and interest rates.
- d) All children under this scheme will be enrolled as a beneficiary under Ayushman Bharat Scheme (PM-JAY) with a health insurance cover of 5 lakhs.

Q.77) Which of the following provisions is incorrect with respect to Pradhan Mantri Swasthya Suraksha Yojana?

- a) Its objective is correcting regional imbalances in the availability of affordable/reliable tertiary healthcare services and also to augment facilities for quality medical education in the country.
- b) Setting up of a new AIIMS (All India Institute of Medical Sciences) is one of its components.
- c) The Ministry of Health and Family Welfare is the nodal agency to implement the scheme.
- d) None of the Above.

Q.78) The "One Stop Centre Scheme" related to?

- a) One stop Vaccination centres at semi urban and rural areas for vaccination of all age groups and

will cater to different types of Covid-19 vaccines and doses.

- b) Providing assistance to Indian women who are survivors of gender-based violence.
- c) Ease of doing business by providing all the license and various authorization related works for starting a business under a single roof.
- d) A single platform where all the information relating to Government services can be accessed and applied as per eligibility norms.

Q.79) Consider the following:

- 1. Principles of Natural Justice
- 2. Powers of a Civil court
- 3. 44th Constitutional Amendment Act
- 4. Chandra Kumar Case, 1997
- 5. Maximum age of chairman and members is 62 years.

Which of the above statements is/are correct regarding the Tribunals under Part XIV-A of the Constitution?

- a) 1, 2, 4, and 5 only
- b) 1, 2, 3, and 4 only
- c) 2, 3, 4, and 5 only
- d) 1, 2 and 4 only

**Q.80) Consider the following pairs:
Infrastructure initiatives**

Associated Countries

- | | | |
|-----------------------------------|---|-----------------------|
| 1. Build Back Better World | - | G-7 |
| 2. Global Infrastructure Facility | - | G-20 |
| 3. Blue Dot Network | - | USA, Japan, Australia |
| 4. Asia-Africa Growth Corridor | - | India-Japan |

Which of the pairs given above is/are correctly matched?

- a) 1, 2, and 3 only
- b) 1, 2, and 4 only
- c) 3 and 4 only
- d) 1, 2, 3, and 4

Q.81) Consider the following statements regarding the Israel-Palestine issue:

- 1. The two-state solution was agreed upon at the Madrid Conference of 1991.
- 2. Dehyphenation policy of India calls for combined talks with the two in case of any project in Jerusalem.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only



- c) Both 1 and 2
- d) Neither 1 nor 2

Q.82) In the context of the Social Stock Exchanges, which of the statements given below is/are correct?

1. It allows both for profit and not for profit organizations to raise funds.
2. India is the first Asian country to come up with this initiative.

Select the correct Answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.83) In the context of Radioisotope Thermoelectric Generators, which of the statements given below is/are correct?

1. It is a nuclear based technology that deals with the generation of power in space.
2. It makes use of the thermocouples and generally it does not involve moving parts.
3. It generates energy due to the natural decay of Plutonium 238.

Select the correct Answer using the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.84) Consider the following statements regarding the Neglected Tropical Diseases (NTDs):

1. Virus, Bacteria, and Helminths are the only causative agents for these diseases.
2. The END7 initiative seeks to end seven Neglected Tropical Diseases by 2022.
3. These are rare diseases found in less than a million of the population in the tropical and subtropical regions.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.85) Consider the following pairs:

Tribes	State/UT
1. Hakki Pikki	- Andhra Pradesh

- 2. Dard - Ladakh
- 3. Angami - Nagaland
- 4. Onge - Assam

Which of the following given above is/are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 and 4 only
- d) 1 and 4 only

Q.86) Consider the following statements regarding Alternative Infrastructure Funds (AIF):

1. Alternative Investment Fund prohibits foreign investors from investing in them.
2. SWAMIH, a kind of AIF-II, is registered under the Securities and Exchange Board of India.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.87) Which of the statements is *least* appropriate regarding the Radio Frequency Identification Technology?

- a) FASTag and E-way bills are integrated with the RFID technology.
- b) It is developed and operated by the National Highway Authority of India (NHAI).
- c) Electronic Product Code (EPC) is a 13-digit code encoded on the FASTag for identification.
- d) Higher security and automatic data logging are its important security features.

Q.88) Consider the following statements regarding the temples situated at Kanchipuram:

1. Kanchipuram is situated on the banks of river Vegavathi.
2. The temples are dedicated to Lord Shiva and Lord Vishnu.
3. Both the Vijayanagar and the Pallava rulers constructed temples on this site.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.89) In India, inclusion of a specific industry in the "Harmonised Master List of Infrastructure sub-



sectors" gives the industry access to cheaper funding including foreign currency funding.

In this regard, which of the following ministries accords the infrastructure status to industries in India?

- a) Ministry of Road transport and highways
- b) Ministry of Home Affairs
- c) Ministry of Finance Affairs
- d) Ministry of Heavy Industries and Public Enterprises

Q.90) Which of the following sites have recently been added to the list of UNESCO World Heritage sites?

- 1. Satpura Tiger Reserve
- 2. Ganga Ghats of Varanasi
- 3. Hirebenakal Megalithic Site in Karnataka

Select the correct Answer using the code given below:

- a) 1 only
- b) 2 only
- c) 3 only
- d) None of the above

Q.91) Which of the following statements with reference to 'New Big 5 Project' is correct?

- a) It is a collaboration between top 5 vaccine manufacturers of the world to increase the production of COVID-19 vaccine.
- b) It is a project to reduce marine plastic in 5 hotspots around the world.
- c) It is a project to protect 5 biggest mountains of the world from human encroachment.
- d) It is a wildlife photography aims to **raise awareness** about the crisis facing the world's wildlife

Q.92) The initiative 'Christchurch Call to Action', recently seen in news, is related to:

- a) Ending the issue of hunger and malnutrition because of COVID-19 pandemic
- b) Providing funding to developing countries to avoid debt-trap diplomacy
- c) Ending online terrorist and violent content
- d) None of the above

Q.93) Consider the following pairs of places in news and their location:

Places in News Country

- 1. Helmand — Afghanistan
- 2. Vorukh — Armenia
- 3. Cali — Colombia

Which of the above pairs is/are correctly matched?

- a) 1 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 3 only

Q.94) Consider the following statements with reference to Putola Nach:

- 1. It is a form of shadow puppetry from Assam.
- 2. The themes include the episodes of Ramayana and Mahabharata along with chants taken from Bhaona.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.95) Consider the following statements about Currency Swap Facility:

- 1. It is a transaction in which two countries exchange an equivalent amount of money but in different currencies.
- 2. It can be used to borrow foreign exchange at lower interest rates.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.96) Consider the following statements about Blackbuck:

- 1. The Velavadar Sanctuary of Gujarat is related to the protected areas of Blackbuck.
- 2. The Bishnoi community considers Blackbuck sacred and worships them.
- 3. In India, Blackbuck is found only in the Deccan Plateau.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above



Q.97) With reference to the State Disaster Response Fund, consider the following statements:

1. Cyclone, Drought, Hailstorm, Landslide, Pest attack and Cold wave are the disasters covered under the State Disaster Response Fund.
2. The Centre contributes 75% of the SDRF allocation for all the states and Union Territories.
3. It is audited by the Comptroller and Auditor General of India every year.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above.

Q.98) In the context of Indian and Oman relations, consider the following statements:

1. Both India and Oman are members of Indian Ocean Rim Association and the Gulf Cooperation Council.
2. Al Nagah is an army exercise conducted between India and Oman.
3. Access to Duqm Port by India is a strengthening factor in India Oman relations.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.99) In the context of the Space Programmes, consider the following statements:

1. Mars Orbiter Mission is the first interplanetary space mission of Indian Space Research Organisation.
2. China's Tianwen-1 became the first spacecraft to land on Mars.
3. OSIRIS-REx is NASA's first mission to visit a near-Earth asteroid.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 only
- c) 3 only
- d) 2 and 3 only

Q.100) Consider the following statements:

1. The "Race To Zero" is a comprehensive energy roadmap released by the International Energy Agency.
2. The "Net Zero by 2050" is a global campaign to rally leadership and support from businesses for a zero carbon recovery that unlocks inclusive and sustainable growth.

Which of the above statements is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the above

Q.1) Ans: A

Exp:

- **Statement 1 is correct:** Area wise **Madhya Pradesh** has the largest **forest cover** in the country followed by Arunachal Pradesh, Chhattisgarh, Odisha and Maharashtra. However, **In terms of forest cover as percentage of total geographical area**, the top five states include **Mizoram** (85.41%), Arunachal Pradesh (79.63%), Meghalaya (76.33%), Manipur (75.46%) and Nagaland (75.31%).
- **Statement 2 is incorrect:** Despite its large swathes of forested land, **the forest survey has found a decrease in the cover in the North East**, to the extent of 765 sq km or 0.45% of the region. **Except Assam and Tripura, all states in the region show decrease in forest cover.** Total forest cover in the North Eastern region is 1,70,541 sq km, which is **65.05% of its geographical area.**

DENSE FOREST (VDF + MDF, insqkm)

FOREST TYPE	2019	2017	CHANGE
Tropical wet evergreen	17,217	17,783	-566
Tropical dry evergreen	565	633	-68
Tropical semi-evergreen	39,596	63,146	-23,350
Tropical moist deciduous	91,399	96,894	-5,495
Tropical thorn	3,765	3,877	-112
Tropical dry deciduous	150,854	146,396	2,458
Littoral and swamp	3,230	4,242	-1,012
Subtropical pine	10,457	11,977	-1,520
Himalayan moist temperate	19,914	21,856	-1,942
Subtropical broad-leaved hill	22,050	10,538	11,512
Subtropical dry evergreen	69	68	1
Montane wet temperate	18,425	3,554	14,871
Himalayan dry temperate	4,077	3,520	557
Sub-alpine	9,740	8,407	1,333
Moist alpine scrub	919	625	294
Dry alpine scrub	712	639	73
Plantation	14,762	12,321	2,441
Total dense forests	407,751	406,476	1,275

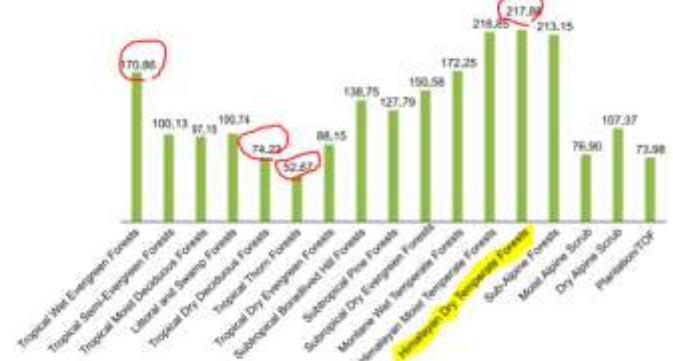
Source: Forest Survey of India

- **Statement 3 is incorrect:**
 - As can be seen in the table here, the **Subtropical dry evergreen forests** have the **least** area (in terms of Sq. km) among the various dense forests recorded in India.
 - They are generally found in the **Bhabar**, the **Shiwaliks** and the **western Himalayas** up to about **1000** metres above sea level.

Q.2) Ans: B

Exp:

FIGURE 9.7 Forest Type wise carbon stock per ha



- According to the '**India State of Forest Report (ISFR), 2019**, The **Himalayan Dry Temperate Forests** have the highest Carbon Stocks per hectare of forests.
 - These are open **evergreen forests with open scrub undergrowth**. Both coniferous and broad-leaved species are present. This type occurs **on the inner ranges** throughout their length and is mainly represented in the north-west.
- Although these forests consist of the highest carbon stocks per hectare, it should be noted that they do not contain the highest carbon stocks in absolute terms. **The tropical Dry deciduous forests consist of the highest carbon stocks in the country - because they occupy the largest absolute area (cover about 65.6 percent of the total forest area).**

Q.3) Ans: C

Exp:

- **Statement 1 is correct**
- - **Tropical Evergreen and Tropical Dry Deciduous forests** coexist in states such as **Maharashtra and Karnataka**. While the **leeward side** of the Western Ghats consists of **tropical dry deciduous forests** and thorn and **Scrub** forests, the **windward side** is primarily made up of Wet Evergreen forests.
- **Statement 2 is correct**
 - States such as **Kerala and West Bengal** consist of both wet evergreen forests and mangrove forest
 - In Bengal, **Tropical evergreen** forests occur in the **terai regions of Darjeeling and**



Jalpaiguri regions ; While the Sunderbans are famous for their Mangroves

- In Kerala, while areas under the **Western ghats constitute the Evergreen forests**, the coastal districts of Kannur, Kozhikode, Ernakulam and Alleppey have limited but important swathes of Mangrove Forests.

• **Statement 3 is correct**

- States such as **Uttarakhand** and **Arunachal Pradesh** consists of both tropical evergreen forests and Alpine vegetation
 - Both **Arunachal Pradesh** and **Uttarakhand** has large swathes of **Tropical Pine forests (evergreen)** in the lower Himalayas and consists of **Alpine vegetation** such as perennial grasses, sedges, cushion plants, mosses, and lichens above the treeline in the upper Himalayas.

• **Statement 4 is incorrect**

As can be seen in this table given by the Forest Survey of India, **Goa, Lakshadweep and Puducherry** have zero or negligible scrub forest vegetation.

2019 Assessment

States/UTs	Geographical area	Very dense forest	Mod. dense forest	Open forest	Total forest	% of geographical area	Scrub
Andhra Pradesh	1,62,968	1,954	13,938	13,206	29,137	17.88	8,255
Arunachal Pradesh	83,743	21,096	30,557	15,036	66,689	79.63	229
Assam	78,438	2,795	10,279	15,253	28,327	36.11	173
Bihar	94,163	333	3,260	3,893	7,306	7.76	250
Chhattisgarh	1,35,192	7,968	32,198	16,345	56,511	41.13	610
Delhi	1,483	6.72	56.42	132.30	195.44	13.18	0.30
Goa	3,702	638	676	1,123	2,237	60.43	0
A & N Islands	8,249	5,678	684	381	6,743	81.74	1
Chandigarh	114	1.36	14.24	6.43	22.03	19.32	0.10
Dadra & Nagar Haveli	451	0	80	127	207	42.16	5
Daman & Diu	111	1.40	5.69	13.40	20.49	18.45	0.19
Lakshadweep	30	0	16.09	11.01	27.10	92.33	0.05
Puducherry	490	0	17.66	34.75	52.41	10.70	0.00

Q.4) Ans: B

Exp:



- The **black soils** are generally **argillaceous (clayey)**, **deep and impermeable**. They **swell and become sticky when wet (rainy season)** and **shrink when dried**. So, during the **dry season**, these soils **develop wide cracks (10-15 cm)**. This facilitates **oxygenation of the soil to sufficient depths**. Thus, **there occurs a kind of self ploughing**. Because of this character of slow absorption and loss of moisture, the **black soil retains the moisture for a very long time**, which **helps the crops, especially the rain-fed ones**, to **sustain even during the dry season**. The soil thus has **extraordinary fertility**.

Q.5) Ans: D

Exp:

- Assessment of biodiversity in forests is important since it provides an indicator to represent the state of conservation of forest ecosystems and it can help to evaluate and monitor sustainability of

the biological resources. It also helps in comparative evaluation of stability, productivity and ecosystem functions of forests in temporal and spatial scales.

- Several quantitative indices have been designed to provide information on different aspects of biodiversity viz, **Margalef index, Menhinick index, Simpson index, Shannon-Wiener Index** etc. The most commonly used index is the **Shannon-Wiener Index** which is based on information theory that provides the biodiversity values and helps to compare it between plant communities/ecosystems. It gives a measure of species abundance and richness to quantify diversity of the species.

Q.6) Ans: D

Exp:

- **Statement 1 is correct and Statement 2 is wrong:**

- Laterite is both a soil and a rock type rich in **iron and aluminum** and is commonly considered to have formed in **hot and wet tropical** areas. Nearly all laterites are of **rusty-red coloration**, because of **high iron oxide content**. They are formed by **intensive and prolonged weathering of the underlying parent rock**.

- Heavy rainfall promotes **leaching (nutrients gets washed away by water)** of soil whereby **lime and silica are leached away** and a soil rich in **oxides of iron and aluminium compounds** is left behind.
- It **lacks fertility** due to a lower **base-exchanging** capacity and a lower content of nitrogen, phosphorus, and potassium. Sometimes, the **phosphate content may be high in the form of iron phosphate**.

- **Statement 3 is incorrect:**

- Continuous stretch of laterite soil is found on the summits of Western Ghats at 1000 to **1500 m above mean sea level, Eastern Ghats, the Rajmahal Hills, Vindhyan, Satpuras and Malwa Plateau**. Continuous stretch of **laterite soil is found on the summits of Western Ghats at 1000 to**

1500 m above mean sea level, Eastern Ghats, the Rajmahal Hills, Vindhyan, Satpuras and Malwa Plateau.

- **Statement 4 is correct:**

- Laterite and lateritic soils provide **valuable building material**. These soils can be easily cut into cakes but hardens like iron when exposed to air. As it is the **end-product of weathering**, it cannot be weathered much further and is durable.

Q.7) Ans: B

Exp:

- **Option 1 is correct:** Alluvial soils are formed mainly due to silt deposited by **Indo-Gangetic-Brahmaputra rivers**. In coastal regions some alluvial deposits are formed due to wave action. Rocks of the Himalayas form the parent material. Thus the parent material of these soils is of transported origin. They are the **largest soil group covering about 15 lakh sq km or about 46 per cent of the total area**. They support more than **40% of India's population** by providing the most productive agricultural lands.
- **Option 2 is incorrect:** These are soils with large amounts of **organic matter** and considerable amount of soluble salts. The most **humid (not arid)** regions have this type of soil. They are black, heavy and highly acidic. Large amounts of carbon, **fixed from the atmosphere into plant tissues through photosynthesis**, are locked away in **peat soils**, representing a **valuable global carbon store**.
- **Option 3 is correct:** After irrigation, the water added to the soil is used by the **crop or evaporates directly from the moist soil**. The salt, however, is **left behind in the soil**. If not removed, it **accumulates** in the soil; this process is called **salinization**. Soil salinization is the key issue in irrigated **arid and semi-arid areas** (as they **facilitate Capillary Action**) that have substantial impact on plant productivity.
- **Option 4 is correct:** These soils occupy about **2.85 lakh sq km or 8.67% of the total land area of India**. They are mainly heterogeneous soils found on the **hill slopes** covered by forests. The formation of these soils is **mainly governed by the characteristic deposition of organic matter** derived from forests and their character changes with parent rocks, ground-

configuration and climate. The **forest soils** are very **rich in humus**. They are **deficient in potash, phosphorus and lime**. They require a **good deal of fertilizers for high yields**.

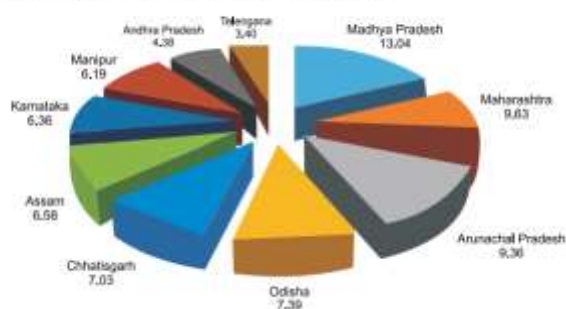
- They are suitable for **plantations of tea, coffee, spices and tropical fruits in peninsular forest region**.

Q.8) Ans: C

Exp:

- India is reportedly home to about **125 indigenous and 11 exotic species of bamboo** from 23 genera. Bamboos occur in abundance in the **deciduous and semi-evergreen forests of the North-eastern region of the country** and the **tropical moist deciduous forests of Northern and Southern India**. The major bamboo genera found in India are **Arundinaria, Bambusa, Chimonobambusa, Dendrocalamus, Dinochloa, Gigantochloa** etc.
- According to the **Indian State of Forest Report 2019**, the **total bamboo bearing area of the country has been estimated to be 16.0 m ha**. **Madhya Pradesh** has a maximum bamboo bearing area of 2.0 m ha followed by Maharashtra (1.54 m ha), Arunachal Pradesh (1.49 m ha) and Odisha (1.18 m ha). As compared to the estimates of ISFR 2017, **the total bamboo bearing area in the country has increased by 0.32 m ha**.

FIGURE 8.2 Top ten States in terms of bamboo bearing area (%)



Q.9) Ans: D

Exp:

All statements are correct:

- Although **unmanned aerial vehicles (UAVs) a.k.a drones** had initially been built for military purposes, they are now widely applied in a number of industries, including agriculture. With the market for agriculture drones reaching **\$1.3 billion**, **agriculture** is now the second largest industry after construction in terms of drone adoption.
 - **Applications:**
 - Smart farming is data-driven, enabling farmers to take action based on accurate information on soil conditions. Equipped with agriculture smart sensors, drones can **collect and deliver this data** – needless to say, they can also do it in a faster and more precise manner.
 - Not only can agriculture drones inform farmers on soil conditions using **thermal, multispectral and hyperspectral technology**, they can also detect field areas inflicted by weeds, infections and pests. **They can locate underperforming areas, enabling better crop management**. Based on this data, farmers can decide on the **exact amounts of chemicals needed to fight infestations**, and not only help **reduce expenses**, but also contribute to better field health.
 - Agricultural fields occupy large areas, and it's often nearly impossible to estimate the overall state of crops. **By using drones for agriculture mapping, farmers can stay updated on the health of plants** in a particular area and indicate which field areas require attention.
 - Remotely accessing fields can help detect bacterial or fungal infections early, and a rapid response to such infections can save an entire crop. **The capability to react quickly allows farmers to apply and monitor remedies** more precisely, ultimately increasing the crop's ability to overcome the disease. All of this improves efficiency and reduces the number and quantity of chemicals released into the environment.
 - The term pesticide treadmill describes the phenomenon whereby pesticide use prevents any natural form of control from taking place, thereby resulting in the need for repeated, routine applications of



pesticide. The pesticide treadmill is said to escalate when the control of

Q.10) Ans: C

Exp:

- **Pesticide Treadmill** one pest provides a window of opportunity for other pests and the need for further chemicals.
 - It is most clearly seen in agricultural operations where farmers become unable to grow their crops without implementing seasonal pest-control programs. An additional complication occurs when similar pesticides are frequently applied: the population of treated pests gradually becomes resistant to the strategies have to be found constantly to control these so-called super pests, and these, in turn, become obsolete with the passage of time.

Q.11) Ans: D

Exp:

- **All options are correct**
 - **Precision agriculture (PA)**, satellite farming is a farming management concept based on **observing, measuring and responding to inter and intra-field variability in crops**. The goal of precision agriculture research is to optimize returns on inputs while preserving resources.
 - **Precision agriculture** has been enabled by advent of **unmanned aerial vehicles, GPS and GNSS, Robotics, Internet of Things and Artificial Intelligence**. The various advancements which have already taken place are:
 1. **Self-steering tractors** have existed for some time now with equipment working like a plane on autopilot. The tractor does most of the work, with the farmer stepping in for emergencies. Technology is advancing towards **driverless machinery programmed by GPS to spread fertilizer or plough land**.
 2. The **Internet of things** is the network of physical objects outfitted with electronics that enable data collection and aggregation. **IoT** comes into play with the **development of sensors and farm-management software**. For example, farmers can spectroscopically measure nitrogen, phosphorus, and potassium in liquid manure, which is notoriously inconsistent.
 - Monitoring technology can be used to make beekeeping more efficient. **Monitoring of a honeybee colony's health via wireless temperature, humidity and CO2 sensors helps to improve** the productivity of bees, and to read early warnings in the data that might threaten the very survival of an entire hive
 3. **Machine learning** is commonly used in conjunction with drones, robots, and internet of things devices. It allows for the input of data from each of these sources. The computer then processes this information and sends the appropriate actions back to these devices. **This allows for robots to deliver the perfect amount of fertilizer or for IoT devices to provide the perfect quantity of water directly to the soil.**
 4. **Telemetry** is the in situ collection of measurements or other data at remote points and their **automatic transmission to receiving equipment**. Products now allow growers to remotely monitor and control nearly every facet of their irrigation operation. The systems save water, time, fuel, and wear and tear on vehicles.
 5. **Smartphones** come with many useful applications already installed, including the camera, microphone, GPS, and accelerometer. There are also applications made dedicated to various agriculture applications such as field mapping, tracking animals, obtaining weather and crop information.

Q.12) Ans: B

Exp:

- **Harrowing and Planking** are examples of **Secondary tillage practices**, carried out to prepare the soil for the ploughing season.
 - **Deep tillage** turns out large sized clods (lump of earth), which are baked by the hot sun when it is done in summer. These clods crumble due to alternate heating and cooling and due to occasional summer showers.
 - **Secondary Tillage** refers to lighter or finer operations performed on the soil after primary tillage. After ploughing, **the fields are left with large clods with some weeds and stubbles partially uprooted.**
 - **Harrowing** is done to a **shallow depth to crush the clods and to uproot the remaining weeds and stubbles.** Disc harrows, cultivators, blade harrows etc., are used for this purpose.
 - **Planking** is done to crush the hard clods to **smoothen the soil surface and to compact the soil lightly.** Thus the field is made ready for sowing after ploughing by harrowing and planking. Generally sowing operations are also included in secondary tillage.

Q.13) Ans: B

Exp:

- **Genetic use restriction technology (Terminator Technology)**
 - Also called the Suicide Seeds Technology, It is the genetic modification of plants to make them produce sterile seeds in second generation. The process involves altering the expression of certain genes in plants so that the plants **terminate their reproductive switch** and make themselves sterile. Such plants that produce seeds that cannot germinate .
 - The development and application of GURTs is primarily an attempt by private sector agricultural breeders to increase the extent of protection on their innovations. There are two types of GURT technologies:
 1. **Variety specific genetic use restriction technologies (V-GURT)**

produce sterile seeds, so the seed from the crop could not be used as seeds, but only for sale as food or fodder

2. **Trait specific genetic use restriction technologies (T-GURT)** are a second type of GURT that would modify a crop in such a way that the genetic enhancement engineered into the crop does not function until the plant is treated with a specific chemical.

Q.14) Ans: D

Exp:

Table 4.6(a): Rice: All-India Area, Production and Yield alongwith coverage under Irrigation

					Area - Million Hectares Production - Million Tonnes Yield - Kg./Hectare
Year	Area	Production	Yield	Area Under Irrigation(%)	
(1)	(2)	(3)	(4)	(5)	
2011-12	44.01	105.30	2393	58.57	
2012-13	42.75	105.23	2461	58.48	
2013-14	44.14	106.65	2416	59.65	
2014-15	44.11	105.48	2391	60.09	
2015-16	43.50	104.41	2400	-	
2016-17	43.99	109.70	2494	-	
2017-18*	43.79	112.91	2578	-	

- **Statement 1 is incorrect:**
 - Area under irrigation of rice has been on the rise, but the trend has been fluctuating at best.
 - Note that, the report here shows the **data from the Agricultural statistics at a glance report, 2018. The 2020 report has not been released by the ministry yet.**

Table 4.7(a): Wheat: All-India Area, Production and Yield alongwith coverage under Irrigation

					Area - Million Hectares Production - Million Tonnes Yield - Kg./Hectare
Year	Area	Production	Yield	Area Under Irrigation (%)	
(1)	(2)	(3)	(4)	(5)	
2011-12	29.86	94.88	3177	93.05	
2012-13	30.00	93.51	3117	93.45	
2013-14	30.47	95.85	3146	93.58	
2014-15	31.47	86.53	2750	94.19	
2015-16	30.42	92.29	3034	-	
2016-17	30.79	98.51	3200	-	
2017-18*	29.58	99.70	3371	-	

- **Statement 2 is incorrect:**



- Wheat, a rabi crop in India, has similarly seen a **fluctuating trend in terms of production.**

Q.15) Ans: D

Exp:

- **Options 1,2,3,5 and 6 are correct**
 - **Mission for Integrated Development of Horticulture (MIDH)** is a Centrally Sponsored Scheme for the holistic growth of the **horticulture sector**. While the **Government of India (GOI) contributes 85%** of total outlay for developmental programmes in all the states except the states in North East and Himalayas, **15% share is contributed by State Governments**. In the case of **North Eastern States** and Himalayan States, GOI contribution is **100%**.
 - MIDH will work closely with the **National Mission on Sustainable Agriculture (NMSA)** towards development of Micro-Irrigation for all horticulture crops and protected cultivation on farmers' fields.
 - It aims to **Promote holistic growth of horticulture sector**, including bamboo and coconut through area based regionally differentiated strategies, which includes research, technology promotion, extension, post harvest management, processing and marketing, in consonance with comparative advantage of each State/region and its diverse agro-climatic features.
 - The various crops/plants/trees covered under the program are:
 - Fruits and vegetables
 - **Root & tuber crops**
 - **Mushrooms**
 - Spices
 - Flowers
 - Aromatic plants
 - **Coconut**
 - **Cashew**
 - **Cocoa**
 - **Bamboo**

Q.16) Ans: A

Exp:

- **Tobacco**
 - In India, **tobacco is grown on 0.45 M ha** of area (accounting for only 0.31% of net cultivated area in the country) with 750 M kg production. The world tobacco production is 7 billion kg, **China occupying the first place** with 2.35 billion kg. There are **ten distinct tobacco types** grown in 15 states of the country which include cigarette (FCV, burley, Oriental) and non-cigarette types (Bidi, chewing, hookah, natu, cheroot, cigar and HDBRG). India stands **second** in tobacco **production** and **third** in **exports** in the world (2020).
 - **Curing** is the term used for **drying tobacco**, and is performed after **harvesting the leaves from the field**. It is a process by which the **harvested tobacco leaf is made ready for the market** (eg: **flue cured virginia tobacco**)
 - Nicotiana Tobacum and Nicotiana Rustica are the important types grown in India.
 - **Conditions for growth**
 - It can withstand a wide range of temperatures varying from **16° to 35°C**.
 - Tobacco needs fairly well distributed rainfall with an **annual range of 50-100 cm**.
 - **Frost** is injurious to its growth. **Bright rainless weather** is helpful at the **curing** stage.
 - **Well drained friable sandy loams** are **ideal** for cultivation. Soils should be **rich in mineral salts** (facilitate full development of roots) but **not in organic matter**.

Q.17) Ans: D

Exp:

- **All Options are correct**
 - **Rabi Crops in India**
 - **Rabi** crops or rabi harvest are agricultural crops that are sown in winter and harvested in the spring in India. The rabi crops are **sown around mid-November**, preferably after the **monsoon**

rains are over, and **harvesting begins in April / May**. The crops are grown either with rainwater that has percolated into the ground or using irrigation. **Good rain in winter spoils the rabi crops** but is good for kharif crops. However, rains such as the **Western Disturbances** in winter prove to be good for these crops

- Major **Rabi Crops** in India are:

Barley	Linseed	Bajra
Gram	Mustard	Lentil
Rapeseed	Wheat	Chickpea

Q.18) Ans: B

Exp:

- According to the latest **Agriculture Stats at a**

Total Nutri Cereals	Rajasthan	6.57
	Karnataka	6.27
	Maharashtra	6.09
	All - India	46.99

Glance Report (2018), Rajasthan is the highest producer of Nutri Cereals - **Sorghum, Pearl Millet, Finger Millet, Foxtail Millet, Bajra, Oats** etc.

- Among other crops, **Rajasthan is the largest producer of barley, mustard, coriander, fenugreek and guar in India**. Rajasthan produces over **72% of guar of the world and 60% of India's barley**. Rajasthan is major producer of **aloe vera, amla, oranges** leading producer of **maize, groundnut**.
- The climate of **Rajasthan is largely arid, showcased best by the presence of Thar Desert**
 - There is very **high variability in the South Western monsoon** in the state. while the **Bay of Bengal branch of monsoon** sheds most of its moisture before reaching Rajasthan, the **Arabian Sea monsoon branch** almost bypasses the state due to the **parallel orientation of the Aravalli ranges**.
 - Another important reason for the large-scale aridity is that the

withdrawal of the monsoon in the first week of September starts from Rajasthan.

- Hence **Rajasthan is the last state where the monsoon reaches And the first State from where the monsoon starts withdrawing**. they are therefore highly dependent on traditional sources of water.
 - Traditional sources of water in Rajasthan include Tanka, Johad, Samund, **Khadeen**, Sarovar, Bawari, Pokhar and **Jhalaras**.

Q.19) Ans: A

Exp:

- Statements 1 is correct**
 - DSR refers to the process of establishing a rice crop from seeds sown in the field rather than by transplanting seedlings from the nursery**. Direct seeding can be done by sowing pre-germinated seed into a puddled soil (wet seeding) or standing water (water seeding) or prepared seedbed (dry seeding).
 - Direct-seeded plants mature **7 to 10 days earlier than transplanted rice**. They are not subjected to stress like being pulled from the soil of the nursery and do not need to reproduce fine rootlets.
 - It offers certain other advantages viz., it saves labour, requires less water, less drudgery, low production cost, better soil physical conditions for following crops and less methane emission, provides better option to be the best fit in different cropping systems
- Statement 2 is correct and 3 is incorrect:**
 - However, despite the obvious upsides, DSR has the following disadvantages:

- There is **greater crop-weed competition** because rice plants and weeds are of similar age. And hence the critical period of **crop weed competition is crucial for highest economic benefits**. In a nursery, weeds do not affect the rice since the environment is controlled.
- The **seed requirement for DSR is also high, 8-10 kg/acre**, compared to 4-5 kg/acre in transplanting.
- Further, laser land levelling is compulsory in DSR. This is not so in transplanting.
- The sowing needs to be done timely so that the plants have come out properly before the monsoon rains arrive.

well in alluvial soils of the Sutlej- Ganga Plain and red and laterite soils of the peninsular region. It **quickly exhausts the fertility of soil**

Q.21) Ans: B

Exp:

- Agribusiness** (bio-enterprise) refers to the enterprises, the industry, the system, and the field of study of the interrelated and interdependent value chains in agriculture and bio-economy. The **primary goal of agribusiness is to maximize profit** while sustainably satisfying the needs of **consumers for products** related to **natural resources** such as **biotechnology, farms, food, forestry, fisheries, fuel, and fiber** — usually with the exclusion of non-renewable resources such as mining. It is also used to describe businesses that are involved in the marketing of farm products, such as warehouses, wholesalers, processors, retailers, and more.

○ **Statement 1 is incorrect:**

- Agri-business is commercial farming on an **industrial scale often financed by corporations ((and not small farmers) with substantial financial Capital** whose main interests lie **outside agriculture**, for example, large corporations in tea plantation business.

- It is not to be confused with **Corporate farming** which is the practice of **large-scale agriculture on farms owned or greatly influenced by large companies**. This includes **corporate ownership of farms and selling of agricultural products**, as well as the roles of these companies in **influencing agricultural education, research, and public policy** through **funding initiatives and lobbying efforts**.

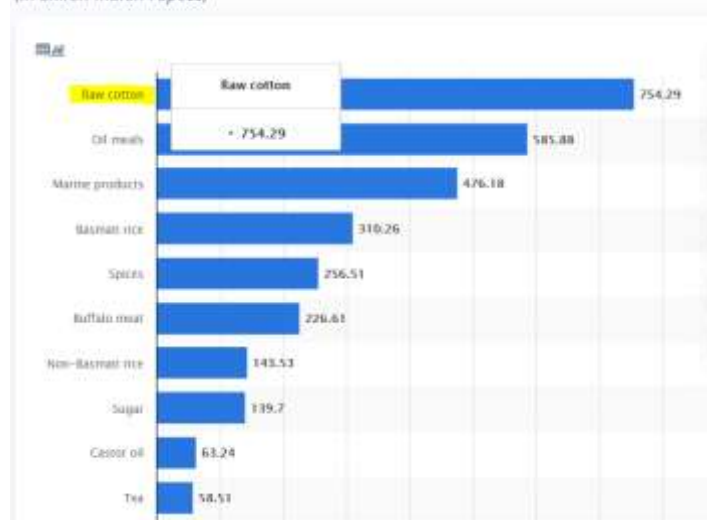
○ **Statement 2 is correct:**

- With **enough access to capital and Technology**, agri businesses are highly **mechanised** with

Q.20) Ans: C

Exp:

Leading agricultural and allied sector exports from India in financial year 2021 commodity
(in billion Indian rupees)



- Cotton**, tropical crop grown in the kharif season in semi-arid areas of the country, made **up the highest value among** agricultural exports from **India in the financial year 2020-21**
- It requires uniformly high temperature (21°C to 30°C), **light rainfall (50-100 cm) or irrigation, 210 frost-free days** and bright sun-shine for its growth.
- Most of the irrigated area under cotton is in **Punjab, Haryana, Gujarat and Rajasthan**.
- Deep black soils (regur-lava soil) of the Deccan Plateau, Malwa Plateau and those of Gujarat** are best suited for cotton cultivation. It also grows

machines such as **Tractors**, drones etc. employed for maximum efficiency and productivity. At the same time, these businesses often rely highly on **chemicals, pesticides and fertilizers** with profit maximization being their primary aim.

too in small quantities. It is **generally absent in the Gondwana rocks**

- **Statement 3 is correct:** Peat contains less than 40 to 55 per cent carbon and has more impurities. It contains sufficient volatile



Q.22) Ans: A

Exp:

- **Statement 1 is incorrect:** Initially peat is converted into lignite or 'brown coal' – these are coal types with **low organic maturity**. In comparison to other coals, **lignite is quite soft and its colour can range from dark black to various shades of brown.**
 - Lignite Contains **40 to 55 percent carbon (higher than Peat but lower than bituminous coal)** and has a **very high moisture** content. Indian lignite deposits occur in the **Tertiary sediments in the southern and western parts of peninsular shield** particularly in **Tamil Nadu, Puducherry, Kerala, Gujarat & Rajasthan and also in Jammu & Kashmir.**
 - **Tertiary Coal** (formed 60 – 15 million years ago) is inferior in quality than **Gondwana Coal** which in turn is inferior in quality than **Carboniferous coal.**
- **Statement 2 is incorrect:** Anthracite coal is the best quality, hard coal with 80 to 95 percent carbon. Various chemical and physical changes cause **Lignite coals to become harder and blacker**, forming the 'bituminous' or 'hard coals'. Under the right conditions and progressive **contrivance of organic maturity**, finally results in the formation of anthracite.
 - Anthracite coal is the best quality, **hard coal with 80 to 95 percent carbon.** It has very little moisture and volatile matter and burns with a short **blue flame (indication of complete combustion)**
 - In India, it is found only in **Jammu and Kashmir** and that

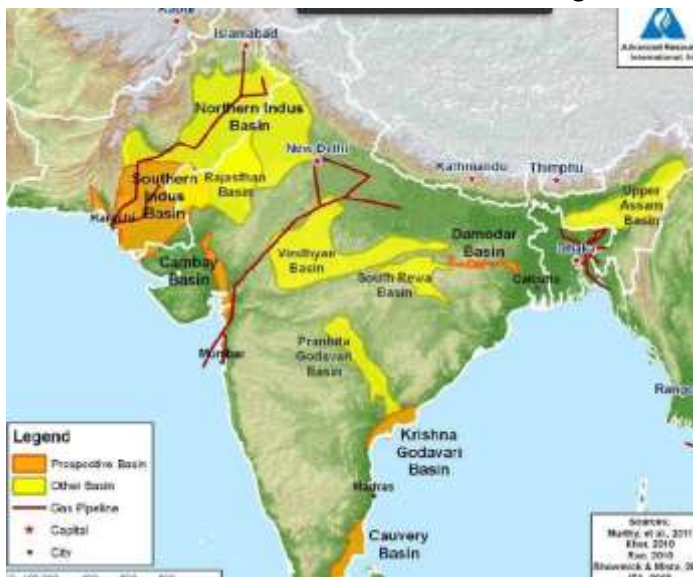
matter and lots of moisture gives out a lot of smoke when burnt and hence is a dangerous pollutant.

- Peat deposits occur in **Nilgiri hills and Kashmir valley** (in the alluvium of the Jhelum).
- **Statement 4 is incorrect:** Gondwana coal contributes an overwhelmingly large proportion of both the **reserves and production of coal, accounting for 98% of the total reserves and 99% of the production of coal in India.** It is the storehouse of India's metallurgical as well as superior quality coal. Out of the **113 major coalfields found all over India, 80 are located in the rock systems of the lower Gondwana age.**
 - It includes coking as well as non coking and bituminous as well as non-bituminous coal. The volatile compounds and ash are present in too large a proportion to allow the carbon percentage to rise above **55 to 60%.** The Gondwana coal is **almost free from moisture** but it contains **Sulphur and phosphorus in small variable quantities.**

Q.23) Ans: D

Exp:

- **Oil Shales** are usually **fine-grained sedimentary rocks** containing relatively large amounts of organic matter from which significant quantities of shale oil and combustible gas can be extracted by destructive distillation.
- Shale gas is a mixture of a lot of **methane, little ethane, propane, & butane, very little carbon dioxide, nitrogen, and hydrogen sulphide.**
- **Sandstone rocks** often have **high permeability**, which means that the tiny pores within the rock are well **connected and gas can flow easily through the rock.** In contrast, **shale rocks** where **gas is trapped as a continuous accumulation** throughout a large area usually have very **low permeability**, making gas production more complex and costly.
- **Shale gas can emerge as an important new source of energy in the country.** India has several Shale Formations which seem to hold shale gas



- The **Ministry of Petroleum and Natural Gas (MoPNG)** has identified six basins as potentially shale gas bearing. These are
 - **Cambay**
 - **Assam-Arakan**
 - **Gondwana**
 - **Krishna-Godavari**
 - **Kaveri**
 - **Indo-Gangetic plain**

Q.24) Ans: D

Exp:

- **All options are correct:**
 - Government has been promoting use of ethanol as a blend stock with main automotive fuel like petrol in line with the **National Policy on Biofuels (NBP)**

-2018 under the Ethanol Blended Petrol (EBP) Programme. This policy envisages an indicative target of blending **20% ethanol in petrol by 2030.**

- Subsequent to amendment in **Industries (Development & Regulation) Act, 1951**, giving **control on production, movement and storage of ethanol to the Central Government**, Central Government has been regularly interacting with the State Governments and other stakeholders to resolve the bottlenecks in **smooth implementation of EBP Programme.**

Category of feed stock	Ex-mill Ethanol Price (Rs./litre)
C-heavy Molasses	45.69
B-heavy Molasses	57.61
Sugarcane juice / sugar / sugar-syrup	62.65
Damaged food grain / Maize	51.55
Surplus rice with Food Corporation of India (FCI) Rice	56.87

- The Government fixes the **ex-mill price of ethanol from sugarcane based raw-materials** and **Oil Marketing Companies (OMCs)** have fixed the price of **ethanol from damaged and surplus rice with FCI**, as shown in the table (for the financial year 2021)

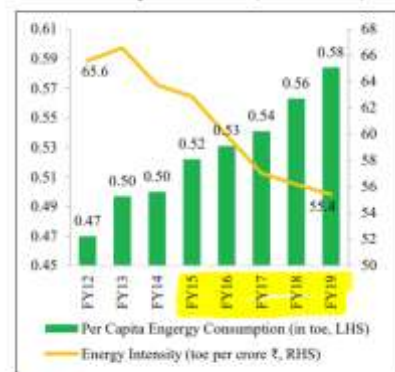
Q.25) Ans: D

Exp:

- **Criticality in Nuclear Power Plants** refer to the normal **operating condition of a reactor**, in which nuclear fuel **sustains a fission chain reaction.** A reactor **achieves criticality** (and is said to be critical) when **each fission event releases a sufficient number of neutrons to sustain an ongoing series of reactions.**
 - **Fission** is a process in which the **nucleus of an atom splits into two or more smaller nuclei, and usually some byproduct particles.** When the **nucleus splits**, the kinetic energy of the fission fragments is transferred to other atoms in

the fuel as heat energy, which is eventually used to produce steam to drive the turbines. For every fission event, if at **least one of the emitted neutrons** on average causes another fission, a **self-sustaining chain reaction** will take place

Figure 42: Energy intensity and Per capita Consumption trend (2011-2019)



Source: Survey calculations based on Ministry Power data.

- **Statements 1 and 2 are correct:**
 - According to the, The Economic Survey 2020-21, Both Per capita Energy Consumption and The Energy Intensity have improved considerably in recent times.
 - **Energy intensity** is defined as the **quantity of energy required to produce a unit of output**. Therefore, **lower the energy intensity the better it is**.
 - The **prime reason** behind such improvement in efficiency is down to the accomplishment of two major landmarks in **rural electrification arena**
 - 100 per cent village electrification under **Deen Dayal Upadhyaya Gram Jyoti Yojana**, and
 - universal household electrification under 'Pradhan Mantri Sahaj Bijli Har Ghar Yojana' (**Saubhaagya**).
- **Statement 3 is correct:**
 - In a significant development in the renewable space, **solar power has overtaken wind power in terms of total installed capacity**.
 - As of January 31, 2021, the total installed capacity of **solar power stood at 38,794 MW** (including 34,561 MW of ground-mounted capacity and 4,233 MW of rooftop capacity). However, the **total wind power capacity was 38,684**



- Recently, The **third unit of the Kakrapar Atomic Power Project (KAPP-3)** in **Gujarat** achieved its 'first criticality.' This is a **landmark event in India's domestic civilian nuclear programme** given that KAPP-3 is the **country's first 700 MWe (megawatt electric) unit**, and the **biggest indigenously developed variant of the Pressurised Heavy Water Reactor (PHWR)**.

Q.26) Ans: D

Exp:

MW, according to data from the **Union Ministry of New and Renewable Energy**

- A decade ago, the solar segment had a capacity of just 18 MW, while wind power's installed capacity was at 13,000 MW.

Q.27) Ans: D

Exp:



- **All options are correct**
- Iron and steel Industries use large quantities of heavy weight losing raw materials and its **localisation is primarily controlled by the availability of raw materials**. Apart from **coal** and **iron** other raw materials such as **limestone, dolomite, Manganese**. All these materials are gross or **weight losing in nature** and therefore the best location for **Iron and Steel Industries is near the source of raw materials**.
 - In India, there is a **crescent shaped region comprising parts of Chhattisgarh, Northern Orissa, Jharkhand and western West Bengal**, which is **extremely rich in high grade iron ore, good quality coking coal** and other supplementing raw materials.

- In recent times however, technological developments in transport, the use of scrap as raw material and agglomeration economics have made **market-oriented** locations more **advantageous** than ever before. **Steel products of an integrated Steel Plant are quite bulky** and it has been estimated that the transport cost per tonne- kilometre of Steel product is about three times more than that of coal or iron ore. Thus, **following the theory of minimum transportation cost many centres of iron and steel production tend to be attracted by the market**.
- **Visvesvaraya iron and steel works at Bhadravathi** is a major exception which is located far away from the main coal producing area of the country. It uses hydroelectricity from **The Sharavathi power project**, unlike other Iron and Steel plants in the **Chhota Nagpur region** which are **dependent on Thermal Power**.

Q.28) Ans: A

Exp:

- **Options 1,2 and 5 are correct:**
 - Footloose industry is a general term for an industry that can be placed and located at any location without effect from factors of production such as resources, land, labour, and capital.
 - These industries often have **spatially fixed costs, which means that the costs of the products do not change despite where the product is assembled**.
 - These industries can be located at a wide variety of places, as these are not **weight-losing** nor **raw-material-specific**. They largely depend on **component parts which can be obtained anywhere**. They produce in **small quantities and also employ a small labour force**. These are **generally not polluting industries**. The **essential factor in their location is accessibility by road network**.
 - Examples include
 - **Diamonds cutting**
 - **Computer chips (Silicon Valley)**



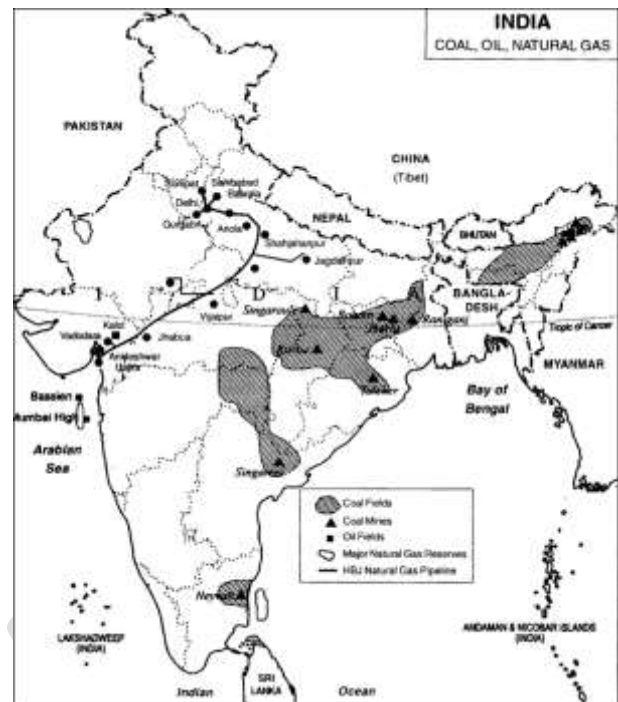
- **Mobile Manufacturing (foxconn, Taiwan)**
 - **Automobile Manufacturing (parts imported from Germany and Assembled in India)**
- **Options 3 and 4 are incorrect:**
- The production of one tonne of **aluminium** requires approximately **6 tonnes of Bauxite and variable quantities of caustic soda lime, fluoride, anthracite**, etc. About 30 to 40% of the production cost of Aluminium is accounted for by electricity alone. This indicates that the **availability of bauxite and electricity** are the two most significant factors which influence the localisation of the industry.
 - Similarly the **localisation of the Fertilizer industry is closely related to the petrochemicals industry**. About **70% of the plants producing nitrogenous fertilizer use naphtha as basic raw material**. That is **why most of the fertilizer plants are located near the oil refineries**. Some fertilizer plants draw that feedstock from **Steel slag** as well as **coke and lignite resources**.
- Q.29) Ans: D
Exp:
- **Statement 1 is correct:**
 - Sugarcane (and also sugar beet) is **bulky, highly perishable and significantly weight losing** (sugar accounts for only ~10% of the bulky sugarcane). Sugar mills can be **operated only during the cane harvesting period**. Hence a **sugar mill must be located close to the cane growing areas**.
 - On the other hand, Sugar Refineries are dependent on brown sugar which is neither perishable nor weight losing. **Sugar refining can therefore be done throughout the year** and hence a sugar refinery can be set up near or **relatively distant location from a sugar mill** - preferably **near ports for ease of exporting**.
- **Statement 2 is correct:**
- Duplication of Pricing of sugar in India has been a major bottleneck in the growth of sugarcane farmers in India. On one hand, the **centre fixes the fair and remunerative price (FRP)** based on the **recommendation of the Commission for Agricultural Costs and Prices (CACP)**. This is governed by the **statutory provisions of the Sugarcane (Control) Order, 1966 issued under the Essential Commodities Act (ECA), 1955**.
 - While on the other hand, the state governments fix **State Advised Prices (SAP)** citing differences in **cost of production, productivity levels** and also as a result of pressure from **farmer groups**.
 - **Dual sugarcane pricing distorts sugarcane and sugar economy** and leads to cane price arrear. **Higher SAP than FRP led to year on year rise in cane price arrears**.
- Q.30) Ans: C
Exp:
- **Options 1,2,3,5 and 6 are correct**
 - Sunrise industry is a colloquial term for a **burgeoning sector or business in its infancy stage showing promise of a rapid boom**. Sunrise industries are typically characterized by high growth rates, numerous start-ups, and an **abundance of venture capital funding**. As a sunrise industry develops, it **may transition to the maturity stage and then to the sunset stage**.
 - **NITI AAYOG** in a paper titled **"Preparing For A Post Pandemic Economy: From SUNSET To SUNRISE Areas Of Growth"** stated that more than half a billion people will live, work and travel in Indian cities in the next decade. This rapid growth will pose several social, economic and environmental challenges. India must take the lead to build new industries that will accelerate

growth and create jobs. Hence the key to a disruptive transformation lies in **five sunrise areas of growth**:

- **Green Mobility** (including hybrid electric vehicles and Green Hydrogen)
- **Advance Cell Chemistries** (EV batteries)
- **Artificial Intelligence (AI)**
- **Genomics**
- **5th generation mobile network technology**
- Other examples of sunrise industries are :
 - **Technical textiles**
 - **Space tourism**
 - **Internet of things**
 - **Blockchain and Fintech**

• **Option 4 is not correct:**

- With the world rapidly marching towards renewable energy to adapt to global warming and climate change, **more and more countries will gradually move away from coal based thermal power plants.** Therefore it could be said that the **thermal power plant technology and industry is in its late mature stage or early sunset stage.**
- Instead, novel Technology such as **solar power, photovoltaic cells, nuclear power etc.** are the **sunrise Industries** today



• **Hajira-Bijapur-Jagdishpur Gas Pipeline (HBJ Pipeline)**

- The project was started in **1986 after the incorporation of GAIL (India) Limited** to supply gas to the fertilizer plants located in the state of Uttar Pradesh. The first phase of the project consisting of a non-



branched **1,750 kilometer grid** was **commissioned in 1987**. Later on the system was expanded with additional branches to supply gas for industrial and domestic use in the states of **Rajasthan, Haryana and NCT**, which increased the total grid length to **3,474 km**.

Q.31) Ans: B
Exp:

Q.32) Ans: C

Exp:

- A special purpose vehicle called the **Dedicated Freight Corridor Corporation of India** is constructing the **Western DFC (approx 1500 km)** and **Eastern DFC (approx 1850 km including Sonnagar-Dankuni PPP Section)**. The EDFC starting from Sahnewal near **Ludhiana (Punjab)** will pass through the states of **Punjab, Haryana, Uttar Pradesh, Bihar and Jharkhand to terminate at Dankuni in West Bengal (6 states)**
 - The **Western Corridor** connecting **Dadri in Uttar Pradesh to Jawaharlal Nehru Port (JNPT) in Mumbai** will traverse through the states of **UP, Haryana, Rajasthan, Gujarat and Maharashtra of WDFC (5 states)**
- Dedicated Freight Corridors** will offer **higher transport output with faster transit of freight trains**. Running of planned double stack container trains and heavy haul will also add to the carrying capacity. The **unit cost of freight transport** will **substantially be reduced** and there will be **significant savings in the Logistics cost** as well. This would improve the supply chain for the industries/logistics players etc. These measures are likely to **create direct and indirect employment in the project-influence areas**.

• **Statement 1 is incorrect:**

- Under **Entry 24 of the Union List** of the Seventh Schedule of the Constitution, the central government can make **laws on shipping and navigation on inland waterways** which are classified as **national waterways by Parliament by law**.
 - Thus, the 111 inland waterways (including 5 National Waterways declared earlier) were declared as **'National Waterways'** under the **National Waterways Act, 2016**.
 - The Act **repeals the five Acts that declare the existing national waterways**. These five national waterways are now covered under the Bill.

• **Statement 2 is correct:**

- The **National Waterway-4** connects the **Puducherry to Godavari River** system via **Krishna River** system **Kakinada** and **Buckingham Canals**. It covers 6 states as shown here.
- The **NW-1 (Ganga-Bhagirathi-Hooghly River System (Haldia - Allahabad))** on the other hand covers **4 states only**.

• **Statement 3 is incorrect:**

Q.33) Ans: B

Exp:

1	National Waterway-1	1620	Ganga-Bhagirathi-Hooghly River System (Haldia - Allahabad)	Uttar Pradesh, Bihar, Jharkhand & West Bengal
2	National Waterway-4	50	Kakinada Canal (Kakinada to Rajahmundry)	Andhra Pradesh, Telangana, Chhattisgarh, Karnataka, Tamil Nadu, Pondichery and Maharashtra
		171	Godavari river (Bhadrachalam to Rajahmundry)	
		139	Eluru Canal (Rajahmundry to Vijayawada)	
		157	Krishna river (Wazirabad to Vijayawada)	
		113	Commamur Canal (Vijayawada to Peddaganjam)	
		316	North Buckingham Canal (Peddaganjam to Central Station of Chennai)	
		110	South Buckingham canal (Central Station of Chennai to Marakkanam)	
		22	Marakkanam to Puducherry through Kaluvelly tank	
		1202	River Godavari (Bhadrachalam - Nashik)	
		636	River Krishna (Wazirabad - Galagali)	



- As of now **National Waterway (NW)-1 Ganga- Bhagirathi- Hooghly river** system from Allahabad to Haldia), **NW-2 (River Brahmaputra from Dhubri to Sadiya)** in Assam, **NW-3 (West Coast Canal** from

Kottapuram to Kollam along with Udyogamandal and Champakara Canals) in Kerala, have already been developed with fairway navigational aids, jetties and terminals with mechanized equipment handling facilities for loading and unloading of cargo. These NWs are operational and vessels are plying on them.

- In addition, **NW-10** (river Amba), **NW-68** (river Mandovi), **NW-73** (river Narmada), **NW-83** (Rajpuri Creek), **NW-85** (Revadanda Creek - Kundalika River System), **NW-91** (Shastri river-Jaigad creek system), **NW- 97** (Sunderbans Waterways), **NW-100** (river Tapi) and **NW-111** (river Zuari) are also operational.
- On the other hand, **Feasibility studies have been completed for 106 new NWs** and based on the outcome of feasibility studies and Detailed Project Reports (DPRs), 20 new NWs have been found technically feasible for development of shipping and navigation in addition to 5 existing NWs by the IWAI.

Q.34) Ans: B

Exp:

- **Statements 1 and 2 are correct:** In a report last year, NITI Aayog said the Railways' share in the transportation of surface freight has declined from 86.2 per cent in 1950-51 to 33 per cent in 2015, owing to a shortfall in carrying capacity and a lack of price competitiveness.
 - The Indian Railways is **12 times more energy efficient in freight traffic** and **three times more in passenger traffic compared to roadways**. But The Road Transport Sector accounts for about **87% of passenger traffic and 60% of freight traffic movement** in the country.
- **Statement 3 is incorrect:**
 - **Roll On Roll Off (RORO)** is a concept of carrying road vehicles loaded with various commodities, on **open flat railway wagons or cargo ships**. In India, RORO services are available with a combination of **roadways and railways and Roadways and waterways**. In India, RORO services

are **provided via both Railways and Waterways**.

- **RO-RO train services** were first introduced in Indian Railways on **Konkan Railways in 1999, and are running successfully** since then.
- Recently, The **RORO vessel named MV Bhupen Hazarika**, built at a cost of Rs 10 crore by the **Inland Waterways Authority of India (IWAI)** with a capacity to carry over 100 passengers along with eight trucks, each weighing 25 tonnes or 25 small vehicles was inducted for over 1.5 lakh residents of **Majuli Island, Assam**.

Q.35) Ans: b

Exp:

- **Statement 1 is Incorrect** : Earthquake swarm is a series of *low magnitude* earthquake occurring in a localised region over days weeks or even months. Earthquakes associated with Volcanic activity often occur in Swarms, Hence these swarms can serve as a marker for the probable presence and hence location of flowing magma associated with the Volcanism. Hence, **Statement 2 is Correct**
- **Statement 3 is Incorrect** : Surface waves are responsible for the most destruction caused by an Earthquake. These are *low frequency, long wavelength* waves of transverse nature i.e the waves travel in direction perpendicular to the direction of propagation of waves

Q.36) Ans: d

Exp:

All statements are Incorrect

- Earthquake Events are scaled either according to Intensity or according to magnitude. Based on these there are 2 scales
- **Richter Scale / Magnitude Scale** : It measures the Magnitude of Energy released during the Earthquake. It is expressed in range 0 - 10
- **Mercalli Scale / Intensity Scale** : Named after Italian Seismologist, It measures the Intensity of visible damage caused by the earthquake. It is expressed in range 1 - 12

Q.37) Ans: b

Exp:

- **Statement 1 is Incorrect** : Bureau of India Standards (BIS) has divided the country into four Seismic Zones : II(Low Intensity) III (Moderate

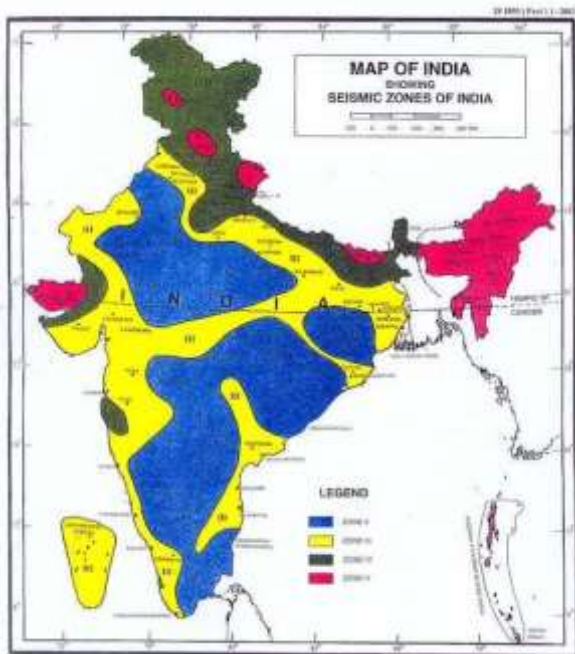
Intensity) IV (Severe Intensity) & V (Very Severe Intensity) : based on the past seismic history

- **Statement 2 is Correct** : Zone V is the most seismically active zone while Zone II is the least seismically active zone
- **Statement 3 is Incorrect** : Area covered under Zones are : Zone II (approx 40%) ; Zone III (approx 30%) ; Zone IV (approx 17.5%) ; Zone V (approx 11%)
- **Statement 4 is Incorrect** : Zones are divided on the basis of *Modified Mercalli Scale* which measures the impact of Earthquake

Q.38) Ans: b

Exp:

- **Only Statement b is Incorrect**
- **Statement a is Correct** : About 60% of India's land area is under threat of moderate to severe earthquakes
- For rest of statements, refer the picture below for identification of different zones and area marked by them



Q.39) Ans: d

Exp:

- **Statement 1 is Correct** : Tsunamis are a series of very long wavelength and period created in ocean by impulsive disturbance such as deep ocean earthquake, undersea volcanoes, Anthropogenic factors such as Nuclear Explosion etc
- **Statement 2 is Correct** : After the December 2004 Tsunami Disaster, India had volunteered to join the International Tsunami warning system

- DOARS (Deep ocean Assessment and Reporting system) was set up in Indian Ocean post 2004
- ITEWC (Indian Tsunami Early warning Centre) embedded with DART (Deep ocean assessment and report of Tsunamis) was established in 2007 at INCOIS

- **Statement 3 is Correct** : The 2004 Tsunami was due to an earthquake of magnitude 9.0 on 26th December in the Indian Ocean Region with its epicentre near the western boundary of Sumatra. The reason of earthquake was that Indian plate went under Burma Plate. The ocean floor was displaced by about 10-20 m with downward direction tilt

Q.40) Ans: a

Exp:

- **Statement 1 is Correct** : Tsunami waves travel at different speeds in water. It behave as shallow water waves because of their long wavelength. They travel at different speed -: slow in shallow water while faster in deep water
- **Systement 2 is Incorrect** : once a Tsunami reaches shallow water near the coast, the wave amplitude increases. The top of the wave moves faster than the bottom, causing the sea dramatically sometimes to height as 20-30 metres above mean sea level

Q.41) Ans: d

Exp:

- **All Statement are Correct**
- Intrusive Landforms are formed by Volcano when Magma cools within the crust
 - Batholith: Cooled down and solidified magma inside the Earth
 - Laccolith: Large dome-shaped intrusive bodies with level base connected by a pipe-like conduit from below.
 - Lapolith: Saucer shaped body concave to the sky.
 - Phacolith: A wavy mass of intrusive rocks found at the base of synclines or at the top of anticline.
 - Sills: Solidified horizontal lava layers inside the Earth.
 - Dykes: Solidified vertical lava layers inside the Earth
- Batholith is a part of original magna chamber which acts as a source for intrusive landforms

- Refer the below figure for better understanding

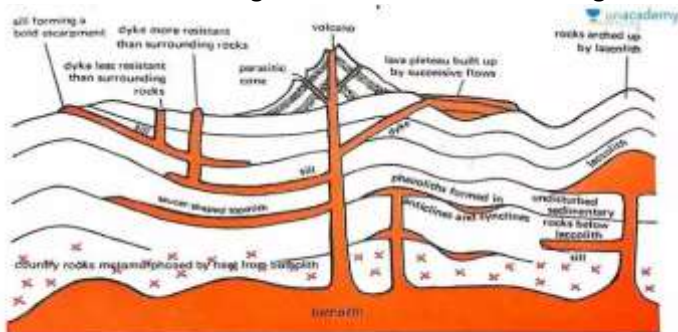


Fig. 26 Intrusive landforms of igneous intrusions in volcanic regions (showing sill, dike, laccolith, lopolith, phacolith and batholith)

Q.42) Ans: a

Exp:

- Statement 1, 2 & 3 are correct**
- Surface water percolates downward through the rocks below the Earth's surface to high-temperature regions surrounding a magma reservoir
 - This water is heated, becomes less dense, and rises back to the surface along fissures and cracks in the form of Geysers, fumaroles (also called solfataras), and hot springs
- Hot Springs** : Springs produced by geothermally heated groundwater. They are very colourful due to presence of cyanobacteria of different colours. They are found all across the world. They have Medicinal value and can be used to harness geothermal energy
 - Example : Tattapaani (Haryana)
- Geysers** : Spring characterised by intermittent discharge of water ejected turbulently. Steam is also a part of ejection. They show distinct colour due to presence of silica deposits. They are found in very few regions.
 - Iceland is famous for its geysers
 - Ex : Old faithful (USA)
- Fumaroles** : They emit mixtures of steam and other gases. These are fed by conduits that pass through the water table before reaching the surface of the ground
 - Example : Black Growler steam vents

Q.43) Ans: d

Exp:

All Statement are Correct

- Hotspot Volcanism** is different from other types as it happened not at the margin of plates but in the Interior parts of Lithospheric plates

- Term "Hotspot" refers to existence of an area inside Earth's Mantle where hot plumes(magma) rises upward through convection and form Volcanoes on underlying rocks. It facilitated the melting of rock at base of Lithosphere.
- Hotspots are relatively fixed in comparison to the Tectonic plates and due to the tectonic plate movement over stationary Hotspot, volcanoes are rafted away and new ones are formed, forming a chain of volcano

Q.44) Ans: a

Exp:

- Statement a is Correct** : Hotspots are relatively fixed in comparison to the Tectonic plates and due to the tectonic plate movement over stationary Hotspot, volcanoes are rafted away and new ones are formed, forming a chain of volcano
- Statement b is Incorrect** : Hawaiian eruption are usually a type of Basaltic Volcanism, hence they are less viscous, more fluid and eruptions are quite while eruption at Mt Vesuvius are Anthracitic type of Volcanoes, hence they are more viscous, less fluid and hence eruption are explosive and with sometimes pyroclasts
- Statement c is Incorrect** : Hotspots can develop beneath Continents. For example : The Yellowstone hotspot, U.S.A
- Statement d is Incorrect** : Hotspot volcanism occurs at abnormally hot centres known as the mantle plume. They are exceptionally hot areas fixed deep below the Earth's crust

Q.45) Ans: c

Exp:

- Statement a is Correct** : Barren Island, lying 135 km north-east of Port Blair became active again in 1991 and 1995. Narcondam Island is probably an extinct volcanic Island.
- Statement b is correct** : Alpidic region or Mediterranean Region houses second most active zone of Volcanic activity after Pacific ring of fire. However, This region also contains one of the most disastrous Volcanism in history like that of Mt Vesuvius
- Statement c is Incorrect** : Ring of fire houses most of the Earth's subduction zone along its stretch
 - But a significant **exception** is the border between the Pacific and North American Plates



- This stretch of the Ring of Fire is a **transform boundary**, where plates move sideways past one another.
 - This type of boundary generates a large number of **earthquakes** as tension in Earth's crust builds up and is released
- **Statement d is Correct** : Scientists have discovered that the youngest parts of the Pacific Plate (about 2 million years old) are cooling off and contracting at a faster rate than older parts of the plate (about 100 million years old). The younger parts of the plate are found in its northern and western parts; the most active parts of the Ring of Fire.

Q.46) Ans: d

Exp:

- **Statement 1 is Incorrect** : Denudational process are Weathering, Mass movements and Erosion/Transportation. Driving force for Weathering are multiple as Gravitational force, Molecular stress, chemical actions etc. For mass movement, Gravitational force is the single driving force while in Erosion, gravity plays no role. It is kinetic Energy driven
- **Statement 2 is Incorrect** : While Weathering mostly is an in situ process means it happens on-site, Mass movement and Erosion or transportation is also an ex-situ process.

Q.47) Ans: c

Exp:

- **Statement 1 is Incorrect** : Exfoliation is a term used to describe the removal of sheets of rocks from a rock's surface due to range of physical and chemical processes during weathering. They form large and *round shaped domes* (Note - A dome can only be round not pointed)
- **Statement 2 is Correct** : Due to Diurnal change in temperature, the surface layers of the rocks tend to expand more than the rock at the depth that leads to formation of stress within the rock. This leads to the surface layer being exfoliated. Since diurnal range of temperature are drastic in Dry climate and High elevation, it is most effective in Dry climate and higher elevation
- **Statement 3 is Incorrect** : A huge rock may undergo disintegration along weak zones. This process known as shattering produces *highly angular pieces* with sharp corners and edges.

Q.48) Ans: c

Exp:

- **Statement 1 is Correct** : weathering plays a significant role in soil formation. Also it aids in soil enrichment as in enhancing certain valuable ores of Iron, manganese, aluminium, copper etc)
- **Statement 2 is Incorrect** : Weathering is not a prerequisite for Mass movement as mass movement can also happen under conditions like removal of support from below, Increase in gradient or height of slopes, overloading through addition of materials. Although weather materials makes it easy for mass movement hence it aids in the process
- **Statement 3 is Correct** : Air, wind, Ice, Heat of Sun etc are Geomorphic Agents. But In mass movement, Gravity is the only driving force. Hence, though Geomorphic Agents aid in mass movement but none are involved in mass movement,

Q.49) Ans: c

Exp:

- **Statement 1 is Correct** : Rock slides and Debris Slides result when rocks or Debris slide down a pre-existing surface. Pile of Talus are common at the base of a rock or a debris slide
- **Statement 2 is Correct** : Both slide and slump are types of Mass movement involving movement of rock mass downslope or altitude under effector gravity. However, slump involves rotation of sliding rock mass along curved surface

Q.50) Ans: d

Exp:

- **Statement 1 is correct** : Glacier landform are associated with formation of U shapes valley. U shaped valley are steep with straight side and flat or rounded bottom. U shaped valley are characteristic of Mountain glaciation in particular
- **Statement 2 is Correct** :
 - Drumlins are elongated, teardrop-shaped hills of rock, sand, and gravel that formed under moving glacial ice. Hence depositional or Aggradational feature
 - A Roche Moutonnée is a rock formation created by the passing of glacier over underlying bedrock which result in asymmetric Erosional form. Hence Erosional or Aggradational feature
 - Drumlins have both upstream and downstream side smooth while Rochee



Mountonne has only upstream side smooth while rough downstream side

- **Statement 3 is Correct** : Glacial erratic is one example where composition of landform won't match with that of the region where it is found. Erratics are stone and rocks that were transported by a glaciers and then left behind after the glacier melted

Q.51) Ans: c

Exp:

- **Statement 1 is correct** : Gorges have equal width at both top and bottom while Canyons have a wider top than its bottom. Both are Erosional landforms by running water.
- **Statement 2 is Correct** : Waterfall formation involves erosion of soft rock more quickly undercutting the hard rock which eventually crashes into the plunge pool, swirl around and cause more erosion. This process repeat itself and waterfall moves upstream. As the waterfall retreats, a steep sides gorge is formed
- **Statement 3 is correct** : Incised meanders occur when a river's base level has fallen giving the river a large amount of vertical erosion power. They are of two types
 - *Entrenched Meanders* : form when river downcuts particularly quickly, due to which there is little opportunity for lateral erosion and hence symmetrical shape
 - *Ingrown Meanders* : Form when river downcuts at less rapid pace giving the river the opportunity to erode laterally as well as vertically
- **Statement 4 is correct** : Alluvial forms is formed when a river flows out of a mountain valley. The river becomes wider and more slower and sediment is deposited in a fan shape. Hence all collection at one place without any Stratification while in Delta, over a long way, coarser material settle out first and finer silt and clay are carried further
- **Statement 5 is Incorrect** : Point bars or meander bars are sediments deposited in *linear fashion* by flowing rivers.
- **Statement 6 is Correct** : In the middle or mature course, River has more energy and a high volume of water. river channel widens and deepens. Large channel means there is less friction , so water flows faster

Q.52) Ans: a

Exp:

- **Statement 1 is Correct** : Karst landform constitute about 8% of Earth's land surface. These are limited to area where comparatively soluble rocks exist , particularly limestone.
- **Statement 2 is Correct** :
 - *Sinkhole* are the collapsed or depressed feature that develop in Karst landscapes. Ground water dissolve the limestone rock below the surface until it eventually becomes unstable and collapses creating local depressional feature
 - *Swallow hole* are cylindrical tunnel like holes lying underneath the sinkhole at some depth
 - Blind valley forms when number of sinkhole coalesce to create a flat floored depression
- When stream comes across these landforms, it gets disappeared and flow internally
- **Statement 3 is Incorrect** :
 - Submerged or Rocky coast have dominant Erosional features. Here the sea is well close to land with a narrow or no coast. most of the west coast of India.
 - Emerged or gentle sloping coast have dominant depositional feature. Here coast is extended by rivers by building coastal planes and deltas
- **Statement 4 is correct** : Tombolo joins two landmass by a connecting bar. It is a depositional marine landform in which an Island is attached to mainland by a narrow piece of land such as spit or bar. It can be called a sandy Isthmus. example - Chesil beach in England.

Q.53) Ans: d

Exp:

- **Pair 1 correct - Red soils** are generally derived from crystalline rock. They develop in a warm, temperate, moist climate under deciduous or mixed forest. The red colour is due to the presence of iron oxide. **They** are poor in lime, magnesia, phosphate, nitrogen and humus. They are fairly rich in potash and potassium.
- **Pair 2 correct - Laterite soils** are formed in hot and wet tropical areas. They develop by intensive and prolonged weathering of the underlying parent rock. High iron oxide content gives it a rusty- red colour. They are rich in bauxite and ferric oxides and sometimes



iron phosphates and poor in lime, magnesia, potash and nitrogen.

- **Pair 3 correct - Black soils (regur)** have rocks of volcanic origin as their parent material. They are typically found in dry and hot regions of the Peninsula. **They** are poor in phosphates, nitrogen and humus. They have a decent amount of alumina, iron oxide, lime and magnesium carbonates.

Q.54) Ans: C

Exp:

- **Statement 1 is incorrect** - The saline-alkaline soils are saturated with salt particles. Due to capillary action or due to evaporation- the salt/mineral rich water when accumulated in the topsoil, leaves behind salt particles.
- **Statement 2 is correct** - In regions with low water tables, salts percolate in subsoil and in regions of good drainage, salts are left behind due to evaporation. This renders the soil after some time, unfit for agriculture as fertility decreases.
- **Statement 3 is correct** - Overirrigation, inundation of sea water in water tables and inland can cause saline-alkaline soils to form.

Q.55) Ans: B

Exp:

Coniferous Forests

Coniferous trees consist of the four major species- Pine, Fir, Spruce and Larch. The trees are conical in shape with sloping branches to adapt to the subarctic climate.

Characteristics

- **Statement 1 is Incorrect:** The leaves are small, thick, leathery and needle-shaped to prevent excessive transpiration.
- **Statement 2 is Correct:** Almost all conifers are evergreen. There is no annual replacement of leaves as in deciduous trees. The same leaf remains on the tree for as long as five years.
- **Statement 3 is Correct:** Coniferous forests are found in climatic regions where altitude reduces the temperature. The conifers are dominant trees in mountainous areas of both temperate and tropical countries.

- Coniferous forests are of **moderate density** and are **more uniform**. **The trees in coniferous forests grow straight and tall.**
- Coniferous forests are also found in regions with high elevation [Example: The forests just below the snowline in the Himalayas].

Q.56) Ans: D

Exp:

- Tropical desert forests are also known as tropical thorn forests. They are found in areas that receive rainfall less than 50 cm and consist of a variety of grasses and shrubs.
- Trees remain leafless for the most part of the year and give an expression of scrub vegetation.
- In India, these forests are found in semi-arid areas of southwest Punjab, Haryana, Rajasthan, Gujarat, Madhya Pradesh, and Uttar Pradesh.
- Important species - babool, ber, date palm, khair, neem, khejri, palas, etc. Hence, option (d) is correct.

Q.57) Ans: B

Exp:

- **Statement 1 is Incorrect** - The laterite soils develop in areas with high temperatures and high rainfall. These are the result of intense leaching due to tropical rains. They develop by intensive and prolonged weathering of the underlying parent rock.
- **Statement 2 is correct** - Due to heavy rain, lime, and silica are leached away, and soils rich in iron oxide, and aluminum compounds are left behind. Humus content of the soil is removed fast by bacteria that thrive well in high temperatures. Laterite soils are **poor** in lime, magnesia, **potash and nitrogen**.
- **Statement 3 is Incorrect** - The laterite soils are commonly found in Karnataka, Kerala, Tamil Nadu, Madhya Pradesh, and the hilly areas of Odisha, and Assam. Whereas the desert soils occupy large areas in the arid parts of western, and north-western India, lying between the river Indus, and the Aravalli mountain range.

Q.58) Ans: B

Exp:

- Saline soils are also known as Usara soils. Saline soils contain a larger proportion of

sodium, potassium, and magnesium, and thus, they are infertile, and do not support any vegetative growth. Their structure ranges from sandy to loamy. They lack nitrogen and calcium.

- **Statement 1 is correct** - Seawater intrusions in the deltas promote the occurrence of saline soils. In the areas of intensive cultivation with excessive use of irrigation, especially in areas of the green revolution,
- The fertile alluvial soils are becoming saline.
- **Statement 2 is incorrect, & Statement 5 is correct** - Excessive irrigation with dry climatic conditions promotes capillary action, which results in the deposition of salt on the top layer of the soil.
- **Statement 3 is incorrect** - Saline soils have more salts, largely because of the dry climate, and poor drainage (In poorly drained soils, most of the pore space is filled with water for long periods, leaving too little air).
- **Statement 4 is correct** - The above discussion can be concluded by saying that Saline soils occur in arid, and semi-arid regions, waterlogged, and swampy areas.

Q.59) Ans: C

Exp:

- Excessive irrigation with dry climatic conditions promotes capillary action, which results in the deposition of salt on the top layer of the soil. In such areas, especially in Punjab, and Haryana, farmers are advised to add gypsum to solve the problem of salinity in the soil. Gypsum improves soils by reducing the amount of sodium present in the soil.
- All the other options promote salinity and are not the consequences of adding Gypsum to the soil. Hence incorrect.

Q.60) Ans: B

Exp:

- **Statement 1 is correct** : Tropical Evergreen forest characterized by dense growth of trees of varying heights, competing for sunlight with climbers and epiphytes. Due to thick growth,

sunlight does not penetrate through and the floor is quite dark.

- **Statement 2 is Incorrect** : The tropical wet evergreen forest in India is usually found in areas receiving **more than 200 cm of rainfall** and having a **temperature of 15-30 degrees Celsius**.
- **Statement 3 is Incorrect** : Tropical Evergreen forests produce valuable hardwood with **greater diversity**. For eg- Mahogany, Rosewood, Ebony, Shisham, bamboo and canes. Due to its very high biodiversity, various patches of tropical evergreen forests have been declared as protected areas to conserve the ecosystem.
- **Statement 4 is correct** : Tropical Evergreen forest found in Andaman and Nicobar, Western slopes of Western ghats, Assam, Meghalaya, Manipur, Mizoram, Tripura.

Q.61) Ans: C

Exp:

- **Pair 1 is correct** : Tropical Thorn Forest includes Babool, ber, wild date palm, khair, neem, khejri, palas etc. Trees are low in height (6 to 10 metres maximum) and widely scattered. They are found in Rajasthan, south-western Punjab, western Haryana, Kachchh and neighbouring parts of Saurashtra.
- **Pair 2 is correct** : Subtropical Broadleaf forests are found in Eastern Himalayas to the east of 88°E longitude at altitudes varying from 1000 to 2000 m. These are forests of evergreen species. Commonly found species are evergreen oaks, chestnuts, ash, beech, sals and pines. They are also found in Nilgiri and Palni hills at higher elevation.
- **Pair 3 is correct** : Moist deciduous forests are found in western ghats surrounding the evergreen forests, Manipur, Mizoram, Odisha, Chota Nagpur Plateau etc. These trees drop their leaves during the spring and early summer when sufficient moisture is not available. **Moist deciduous forests** occupy a **much larger area than the evergreen forests**. **These provide valuable timber like Teak.** The main species found in these forests are teak,



sal, laurel, rosewood, amla, jamun, bamboo, etc.

- **Pair 4 is Incorrect** : Tropical Evergreen Forests do not shed their leaves together. From the air, the tropical rain forest appears like a thick canopy of foliage, broken only where it is crossed by large rivers or cleared for cultivation. It can be found in - Western side of the Western Ghats, Some regions in the Purvanchal hills and Andaman and Nicobar Islands. The important species of Evergreen forests are mahogany, mesua, white cedar, jamun, canes, bamboo etc. **Teak is valuable timber of Moist deciduous forests**

Q.62) Ans: C

Exp:

- **Statement 1 is correct** : The **Southern slopes in Himalayan region are covered with thick vegetation**, whereas the northern slopes are generally barren. Southern slopes receive more precipitation and vertical rays during the middle of the day. While northern slopes are in a rain shadow area and usually receive sun rays only for a few hours during the day at a low angle.
- **Statement 2 is Incorrect** : **Eastern Himalaya is much** ahead from western Himalayas in terms of Biodiversity and is one of the Biodiversity hotspots. **Eastern Himalayas** is a part of the **Himalayan** global **biodiversity** hotspot. This region is exceptionally rich in diversity and endemism.
- **Statement 3 is Incorrect** : The **Shola vegetation** are tropical montane forests (Temperate) found in the Western Ghats (Nilgiris, Anaimalai and Palani Hills) separated by rolling grasslands at high **altitudes**. Shola grasslands include dwarf trees growing 25-30 feet. It is a stunted forest growth of diverse grass species.

Q.63) Ans: c

Exp:

- **Statement 1 is correct**: The move was approved by the cabinet in October 2020 and is likely to benefit around 3.7 lakh workers

dependent on the jute sector. The government is required to consider and provide for the compulsory use of jute packaging material.

- **Statement 2 is incorrect**: It thrives in tropical LOWLANDS areas with 60-90 percent humidity. It is a rainfed crop with little need of fertilizer and pesticide.
- **Statement 3 is correct**: The cultivation of jute in India is confined to the eastern region of the country. Yearly flooding and good rainfall provides good natural conditions for growth of the plant. The first jute mill was established in Rishra (WB) in the year 1855.

Q.64) Ans: C

Exp:

- **Statement 1 is correct**: Hybrid seeds can give better yield but it is impractical to save them to be used in the next season. They may display different characteristics than what you expect. In the case of Suvin they can be reliably saved. It was created by crossing Egyptian cotton variety "Sujata" with "St Vincent" from the West Indies. In the 1980s, India was producing around 35,000 to 40,000 bales of Suvin, whereas the current production is just around 1,500 bales. In Japan this variety is termed as SUVIN gold.
- **Statement 2 is correct**: India implemented a technical assistance programme (TAP) for cotton in six African countries, namely Benin, Burkina Faso, Chad, Malawi, Nigeria and Uganda, from 2012 to 2018.
- In the five year long second phase, the programme will be scaled up in size and coverage and will be introduced in five additional countries, namely Mali, Ghana, Togo, Zambia and Tanzania. The Cotton TAP programme will now cover 11 African countries including the C4 (Benin, Burkina Faso, Chad and Mali)

Extra edge:

- India is the largest producer of cotton (FAO 2020). But this is due to the increased acreage over the years and not due to

productivity. Cotton occupies around 5% of the total cropped area in the country.

- India is believed to be the original home of cotton plants. It is a kharif crop and requires 6-8 months to mature. In India, Gujarat, Maharashtra and Telangana are the leading producers.
- India grows both short (Indian) as well as long stapled (American) cotton.

Q.65) Ans: C

Exp:

- **Statement 1 is Incorrect.** Considerable quantities of Coal Bed Methane (CBM) are trapped as free gas in the joints and fractures of the coal seam. Large quantities of gas are adsorbed on the internal surfaces of the micropores within the coal itself. This gas can be **accessed by drilling wells into the coal seam and pumping large quantities of water that saturate the seam.** water will occupy the gaps and pores and will push out the gas
- **Statement 2 is correct :** Shale gas occurs frequently at depths exceeding 1,500 metres. Extraction is done through **horizontal drilling** through the shale seam, followed by **hydraulic fracturing, or fracking**, of the rock by the injecting of fluid at extremely high pressure.
- **Statement 3 is correct.** Adding guar gum **increases viscosity of water and makes high-pressure pumping and the fracturing process more efficient.** High viscosity water is much more effective at suspending sand grains and carrying them into the fractures. The guar bean is grown mainly by farmers in **Rajasthan and Haryana.**

Extra Edge by Only IAS

Hydraulic Fracking

- Shale rock is sometimes found 3000-4000 metres below the surface.
- After deep vertical drilling, there are techniques to drill horizontally for considerable distances to extract the gas-rich shale.
- A mixture of water, chemicals, and sand is then injected into the well at very high pressures to create a number of fissures (Cracks) in the rock to release the gas.
- The process of using water for breaking up the rock is known as '**hydro-fracturing**' or '**fracking**'.
- The chemicals help in water and gas flow and tiny particles of sand enter the fissures to keep them open and allow the gas to flow to the surface.

Q.66) Ans: D

Exp:

- **Statement 1 is Incorrect : Photovoltaic (PV) panels turn sunlight (Solar energy) into electrical energy.** They're made from a semiconductor like silicon, in two layers to produce an electric field. When sunlight strikes this field, it generates a small voltage. The panels send this direct current (**DC**) to an inverter, which turns it into alternating current (AC) - used by appliances in the home. Electricity flows from the fuse box into these appliances if required, or out into the grid if not.
- **Statement 2 is Incorrect : Solar thermal panels turn sunlight into heat.** The panel is composed of tubes filled with a liquid. Sun heats the liquid, which is then sent to heat up a copper coil and warm your hot water tank. Solar Roof top water heater utilises solar energy to heat water and if the same energy is utilised to make steam which is further used to rotate turbines and generate electricity.

Q.67) Ans: A



Exp:

- **Statement 1 is Incorrect** - Major minerals are those specified in the first schedule appended in the Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act 1957) and the common major minerals are Lignite, Coal, Uranium, iron ore, gold etc.
- **Statement 3 is correct** - India's more than 90 % Coal Reserve and Production comes from Gondwana coal fields. India is the world's second largest coal producer and the 5th largest country in terms of coal deposits. Jharkhand, Odisha, Chhattisgarh are leading coal producing states in India.

Q.68) Ans: D

Exp:

- **Statement 1 is Incorrect** : India has no significant reserves of Uranium. All needs are met through imports. India imports uranium from Russia, Kazakhstan, France. India has made several bilateral arrangements to meet her demand of Uranium for peaceful energy production as India is **not a member of the Nuclear Proliferation treaty**. Some quality reserves were recently discovered in parts of Andhra Pradesh and Telangana between Seshachalam forest and Sresailam.
- **Statement 2 is Incorrect** : Thorium is derived from monazite sand. Thorium deposits are found in Kerala, Jharkhand, Bihar, Tamil Nadu, and Rajasthan. But the largest concentration of **monazite** sand is on the **Kerala coast**. Over 15,200 tonnes of uranium and Thorium is estimated to be contained in monazite.
- **Thorium** cannot in itself **power** a **reactor**; unlike natural uranium, it does **not** contain enough fissile material to initiate a **nuclear** chain reaction. Thus the Thorium based nuclear reactor forms the third stage of India's Three stage nuclear power programme.
- **Statement 3 is Incorrect**. Beryllium oxide is used as a '**moderator**' in nuclear reactors of power plants. India has sufficient reserves of beryllium to meet its requirement of atomic power generation. Its reserves are in the states

of Rajasthan, Jharkhand, Chhattisgarh, Andhra Pradesh and Tamil Nadu.

Q.69) Ans: C

Exp:

- **Statement 1 is Incorrect** - The 'Lithium Triangle' is a region of the Andes which is rich in Lithium reserves around the borders of **Argentina, Bolivia and Chile. It accounts for 54% of world lithium resources.**
- **Statement 2 is Incorrect** - **China** is the world's largest **consumer of lithium**, due to the size of its battery-manufacturing industry for electric vehicles and consumer electronics. **India currently imports all its lithium needs to meet its domestic demands.**
- **Statement 3 is correct** - Preliminary surveys by the Atomic Minerals Directorate for Exploration and Research (AMD), an arm of the Department of Atomic Energy, are learnt to have shown the presence of 1,600 tonnes of lithium resources in the **Margalla-Allapatna area**, along the Nagamangala Schist Belt, which exposes mineralised complex pegmatites (igneous rocks).

Q.70) Ans: C

Exp:

- **Statement 1 is Incorrect** : According to the Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act 1957), the **Central government** notifies minor minerals through notification. State government has power to control illegal mining and also power to frame rules for that.
- **Statement 2 is correct** : Today, National Mineral Development corporation (NMDC) is the only organised producer of diamonds in India from its **Majhgawan mine at Panna, Madhya Pradesh**. By states, Madhya Pradesh accounts for about 90.20% resources followed by Andhra Pradesh 5.71% and Chhattisgarh 4.08%.
- **Statement 3 is correct**. **Karnataka is the largest producer of gold** in India. Gold mines are located in Kolar [**Kolar Gold Field**], Dharwad, Hassan and Raichur [**Hutti Gold**]

Field] districts. Kolar Gold Fields is one of the deepest mines of the world.

- **Andhra Pradesh is the Second largest producer of gold in India.** Ramagiri in Anantapur district is the most important gold field in AP.

Q.71) Ans: D

Exp:

There are 40 UNESCO World Heritage Sites in India.

Dholavira and Ramappa Temple are the latest additions to the list under the 'Cultural' category. The other categories — Natural and Mixed.

- **Statement 1 is correct:** All the Natural and Mixed sites are under protection of Ministry of Environment, Forest and Climate Changes
- **Statement 2 is correct:** Historic City of Ahmedabad (2017) is under Protection of Ahmedabad Municipal Corporation. Victorian and Art Deco Ensemble of Mumbai (2018) is Under protection of Bombay Municipal Corporation. Jaipur City, Rajasthan (2019) is under Protection of Jaipur Municipal Corporation
- **Statement 3 is correct:** All the Cultural sites are under protection of the Archaeological Survey of India. The Archaeological Survey of India (ASI), works as an attached office under the Department of Culture, Ministry of Tourism and Culture, is the premier organization for the archaeological research and protection of the cultural heritage of the nation.
- **Statement 4 is correct:** Mountain Railways of India Darjeeling,(1999)- West Bengal, Nilgiri (2005)- Tamil Nadu, Kalka-Shimla (2008) - Himachal Pradesh and Chhatrapati Shivaji Terminus (formerly Victoria Terminus) (2004) - Maharashtra are under protection of Ministry of Railways.

Q.72) Ans: B

Exp:

- **Statement 1 is incorrect :** Role of RO is mentioned in Part IV of Representation of People Act 1951. The returning officer of an assembly constituency performs statutory functions under the Representation of the People Act, 1951 in quasi-judicial capacity independently
- **Statement 2 is correct :** The EC can designate or nominate the same person to be returning officer for more than one constituency. The EC appoints

the Returning Officer and Assistant Returning Officer for a constituency in consultation with the governments of the State or Union Territory as the case may be.

- **Statement 3 is correct :** The Returning Officer's duties include accepting and scrutinizing nomination forms, publishing the affidavits of candidates, allotting symbols to the contesting candidates, preparing the list of contesting candidates, preparing the EVMs and VVPATs, training polling personnel, designating counting centres, and counting the votes and declaring the result.

Q.73) Ans: D

Exp:

- The Centre has no role to play in the alteration of districts or creation of new ones. States are free to decide on this matter. The power to create new districts or alter or abolish existing districts rests with the State governments. This can either be done through an executive order or by passing a law in the State Assembly. Know India, a website run by the Government of India, says currently there are 718 districts in the country, (According to the 2011 Census, there were 593 districts in the country).
- The Home Ministry comes into the picture when a State wants to change the name of a district or a railway station. The State government's request is sent to other departments and agencies such as the Ministry of Earth Sciences, Intelligence Bureau, Department of Posts, Geographical Survey of India Sciences and the Railway Ministry seeking clearance. A no-objection certificate may be issued after examining their replies.

Q.74) Ans: B

Exp:

- **Statement 1 is correct:** The format was founded in 2012 in Budapest **to push for the cooperation of the "17+1" (the 17 CEE countries and China).** The 17+1 (China and Central and Eastern Europe Countries) initiative is a China-led format founded in 2012 in Budapest with an aim to expand cooperation between Beijing and the CEE (Central and Eastern Europe) member countries, with investments and trade for the development of the CEE region.
- **Statement 2 is incorrect: The initiative includes members from five Balkan countries and twelve**



EU member states, namely Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Estonia, Greece, Hungary, Latvia, Lithuania, Macedonia, Montenegro, Poland, Romania, Serbia, Slovakia and Slovenia.

- Recently Lithuania quit China's 17+1 cooperation forum with central and eastern Europe, Lithuania has called it "divisive" and urged fellow EU members to pursue "a much more effective 27+1 approach and communication with China."

Q.75) Ans: B

Exp:

- The Arctic Council is the leading intergovernmental forum promoting cooperation, coordination and interaction among the Arctic States, Arctic Indigenous peoples and other Arctic inhabitants on common Arctic issues, in particular on issues of sustainable development and environmental protection in the Arctic. It was **formally established in 1996 by the Ottawa declaration**. ARCTIC STATES : Canada, Denmark, Finland, Iceland, Norway, Russia, Sweden, and the United States.
- The Arctic Council does not prohibit the commercial exploitation of resources in the Arctic. It only seeks to ensure that it is done in a sustainable manner without harming the interests of local populations and in conformity with the local environment. India is an Observer State of the Arctic Council.

Q.76) Ans: C

Exp:

- The scheme has been launched for support & empowerment of Covid affected children. Eligibility: All children who have lost both parents or surviving parent or legal guardian/adoptive parents due to Covid 19 will be supported under the scheme.
- Option C is incorrect:** The child will be assisted in obtaining education loan for Professional courses / Higher Education in India as per the existing Education Loan norms. The interest on this loan will be paid by the PM CARES FUND.
- As an alternative, scholarships equivalent to the tuition fees / course fees for undergraduate/ vocational courses as per Government norms will be provided to such children under Central or State Government Schemes. For children who are not eligible under the existing scholarship

schemes, PM CARES will provide an equivalent scholarship.

Q.77) Ans: D

Exp:

- Pradhan Mantri Swasthya Suraksha Yojana (PMSSY), a Central Sector Scheme, was announced in August 2003 to address imbalances in availability of tertiary care hospitals and improve medical education in the country.
- The Scheme two components, namely;
 - Setting up a new AIIMS (All India Institute of Medical Sciences).
 - Upgradation of government medical colleges in various states.

The project cost for upgradation of each medical college institution is shared by the Centre and the state.

So far the setting up of 22 new AIIMS has been approved under the scheme, out of which six AIIMS at Bhopal, Bhubaneswar, Jodhpur, Patna, Raipur and Rishikesh are already fully functional.

- The PMSSY is implemented by the Ministry of Health and Family Welfare.

Q.78) Ans: B

Exp:

- One Stop Centre Scheme (OSCs) being implemented by the Ministry of Women and Child Development and has provided assistance to over 3 lakh women so far.
- The scheme is being implemented across the country since 1st April, 2015 through State Governments/ Union Territory (UT) Administrations to provide **integrated support and assistance to women affected by violence and in distress**, both in private and public spaces, under one roof and facilitate immediate, emergency and non-emergency access to a range of services including police, medical, legal aid and counseling, psychological support to fight against any forms of violence against women. Till date, 701 OSCs in 35 States/UTs have been operationalised.
- Recently, The Central government announced to set up One Stop Centres (OSCs) across 10 missions to provide assistance to Indian women who are survivors of gender-based violence. The missions where the OSCs will come up are Bahrain, Kuwait, Qatar, Oman, UAE, Jeddah and



Riyadh in Saudi Arabia, Australia, Canada and Singapore.

Q.79) Ans: D

Exp:

- **Article XIV-A** of the **Constitution of India** was added by the **42nd Constitutional Amendment Act, 1976** and provided for the constitution of **Tribunals**. It contains two articles- **Article 323 A** dealing with the administrative tribunal and **Article 323 B** dealing with the tribunals for other matters. **Article 323 A** of the Constitution empowers the Parliament to constitute a tribunal for the adjudication of disputes regarding the service matters and unburden the civil and the high courts. In pursuance of this, the Administrative Tribunals Act, 1985 was enacted by the Parliament that empowered the Central government to constitute the Central Administrative Tribunal and the State Administrative Tribunals.
- The tribunals have the **powers of the civil courts** and work on the principle of **Natural Justice**. It is not bound by the procedure laid down under the Civil Procedure Code of 1908. The Central Administrative Tribunal is constituted as a multi-member body with a chairman and the members. The **retirement age of the chairman is 65 years while that of the members is 62 years**. The appointment is done by the Appointment committee of the Cabinet which is selected by the high-powered search committee composed of the judges of the supreme court nominated by the Chief Justice of India. The original Act allowed the appeals from the tribunals to be done to the supreme court only and not to the High Courts. However, the supreme court in the **Chandra Kumar case, 1997** provided that the appeals could be made even to the concerned high court as well and not just to the supreme court.

Q.80) Ans: D

Exp:

- **Build Back Better World** partnership was launched by the **G-7 countries** as a global infrastructure project led by the major

democracies to channelize 40 trillion dollars for the developing countries post the COVID pandemic.

- **The Global Infrastructure Facility** is an initiative of the **G-20 countries** to boost private investments in infrastructure projects in developing and emerging countries.
- **Blue Dot network** certifies infrastructure projects in the Indo-Pacific regions and makes them eligible for funding. It channels finance from the financial institutions of the **USA, Japan, and Australia**.
- **The Asia-Africa Economic Corridor** aims to promote **Indo-Japan** collaboration to promote digital infrastructure and connectivity in the African region. It is an economic partnership between the two.

Q.81) Ans: A

Exp:

- **Statement 1 is correct:** Recent clashes between Israel and Palestine in the Gaza strip were calmed with the mediation led by Egypt. Hamas, a Palestinian faction controls the Gaza strip and the conflict between the two is on account of the control over Jerusalem, a city considered holy by the Islamic, Judaism, and the Christians. India has always favoured the cause of Palestine. However, in the **Madrid peace conference, 1991** the **two state solution** was agreed upon and India agreed to it. India established full diplomatic relations with Israel in 1992. However, it did not give up its stand on Palestine.
- **Statement 2 is incorrect:** India came up with the **dehyphenation policy** in the context of relations between **Israel and Palestine** in 2018. As per this policy, the relation of India with Israel would be totally independent of Palestine and would have its own merits. It would in no way be influenced by the circumstances and its stance with Palestine.

Q.82) Ans: A

Exp:

- **Statement 1 is correct:** The idea of **Social Stock Exchanges** was put forth in the Budget 2019 and consequently a working group was constituted by SEBI to recommend the functioning of these stock

exchanges. Social stock exchanges are platforms that enable For the Profit and Not for Profit organizations to raise finance for their economic, social, environmental, and other similar projects. Earlier, these organizations would raise finance through the grants by the governments, donations, gifts, Corporate Social Responsibilities, impact investors, etc. However, this would open new channels for their funding.

- **Statement 2 is incorrect:** India is not the first country in Asia that has come up with this. UK, Canada, South Africa already possess an arrangement of this kind. Besides, Singapore in Asia also has a similar arrangement.

Helminths (Lymphatic filariasis, River Blindness, Soil transmitted Helminthiasis, etc.).

- **Statement 2 is correct:** END7, an international public awareness campaign to end the seven neglected tropical diseases (NTDs) by 2020. The Seven NTDs are lymphatic filariasis (elephantiasis), river blindness (onchocerciasis), snail fever (schistosomiasis), trachoma, hookworm, whipworm (trichuriasis) and roundworm (ascariasis). The above 7 diseases account for around 90% of the total burden of the NTDs. WHO's strategic roadmap for 2021-2030 also aims to make efforts to end NTDs.
- **Statement 3 is incorrect:** They are **communicable diseases** that are found in the tropical and the sub-tropical countries and are found to affect greater than one billion people around the world. Populations suffering from high incidences of poverty, unhygienic conditions of living, and lack of sanitation are at a greater risk of the miseries caused by these diseases.

Q.83) Ans: D

Exp:

- **Statement 1 is correct:** **Radioisotope Thermoelectric Generator** or RTGs is a nuclear energy based technology that aims to convert Thermal energy into Electric energy and is used for the generation of power and management of heat in various space missions. Recently, ISRO's U. R. Rao Satellite Centre invited proposals for the three phase development of RTGs for its deep space missions.
- **Statement 2 is correct:** An advantage of RTGs is the use of the **Thermocouple**. The generators are governed by the "Seebeck effect" and it generally has no movable parts. Seebeck effect is a phenomenon in which a temperature difference between two dissimilar electrical conductors or semiconductors produces an electromotive force (electricity).
- **Statement 3 is correct:** RTGs make use of the natural decay of **Plutonium-238** for the generation of power. The heat produced by its decomposition is transferred to the thermocouple which is responsible for the generation of electricity.

Q.85) Ans: B

Exp:

- The **Hakki Pikki** tribe is mainly found in the state of **Karnataka**. They migrated from the northern states and now settled in southern India. It is a nomadic tribe and is known as a bird catcher. They speak the *Vaagri* language which is the Indo-Aryan language categorized as Endangered by UNESCO.
- **Dard Tribes** are the **Buddhist tribes** belonging to **Ladakh** and are mainly found in the districts of Leh and Kargil. These tribes belong to the Aryan culture and the area inhabited by them is referred to as the Aryan Valley.
- **Angami tribes** are the hill tribal groups of the state of **Nagaland** and are also found in some parts of Manipur. They are settled in the hill regions and are known for wet-rice cultivation.
- The **Onge tribe** belonged to the traditional hunter-gatherer community in the **Andamans**. The Great Andamanese, Jarawa, Sentinelese Onge and Shompen are recognized as Particularly Vulnerable Tribal Groups.

Q.84) Ans: C

Exp:

- **Statement 1 is incorrect:** **Neglected Tropical Diseases** are caused by the **Viruses** (Dengue, Rabies, etc.), **Bacteria** (Leprosy, Trachoma, Yaws, etc.), **Protozoa** (Chagas, Leishmaniasis, etc.), and

Q.86) Ans: A

Exp:



- **Statement 1 is correct:** **Alternative Investment Funds** are privately pooled investment funds that allow investment from both domestic and foreign investors. It is established in India and categorized into three categories.
 - **Alternative Investment Funds I** are the ones that invest in the startups and the early ventures or the projects that are regarded as socially or economically important by the government.
 - **Alternative Investment Funds II** deals with real estate and the funds for distressed assets, etc.
 - **Alternative Investment Funds III** deals with different types of funds like hedge funds, etc.
- **Statement 2 is incorrect:** **SWAMIH or the Special Window for Affordable and Mid Income Housing Fund** was created by the government in 2019 to provide priority funding to the stalled housing projects for mid income housing. This is a SEBI registered fund under the category of AIF II.

Q.87) Ans: B

Exp:

- **Radio Frequency Identification technology** is a wireless technology that works on the radio waves. It makes use of the electronic tag and an RFID reader that is used for the purpose of identification of the tag. This technology finds its application in the collection of tolls from the vehicles running in the highways.
- A **FASTag** is placed on the vehicle connected with the bank account of the person, and is used for the automatic collection of tolls in the vehicles. The FASTag, when it comes in the vicinity of the RFID reader by making the use of the radio waves, identifies the particular vehicle using the 13 digit Electronic product code unique for each vehicle.
- The **FASTag** is developed by the **National Payment Corporation of India** and is operated by the **National Highway Authority of India**. It offers a number of advantages like the seamless movement of the traffic due to automatic collection of toll, automatic scanning and data logging, and higher security as it is difficult to counterfeit.
- Recently, E-way bills were also integrated with FASTag and RFID. E-way bill is an electronic document to be carried by a person carrying a

consignment of goods with a value greater than Rs. 50000 whether inter or intra state. It is mandated by the GST.

Q.88) Ans: D

Exp:

- **Statement 1 is correct:** **Kanchipuram** is situated on the banks of **river Vegavathi** in the present day state of Tamil Nadu. It was home to around 1000 temples while 126 are remaining at present. Kanchipuram is also famous for its silk weaving work.
- **Statement 2 is correct:** Out of the 126 temples remaining in the city 108 temples are dedicated to Shiva while the 18 temples are dedicated to **Vishnu**. **Kailasanathar** and the **Ekambareshwar Temple** are dedicated to Lord Shiva while the **Varadharaja Perumal Temple** is dedicated to Lord Vishnu.
- **Statement 3 is correct:** **Kanchipuram** was the capital of the **Pallava Kingdom** from the 6th to 9th Century. It was famous for its Dravidian style of temple architecture. **Kailashnathar Temple** was built by the Pallava ruler Rajasimha I and is marked by the 16 sided shivling. **Vijayanagara ruler Krishnadevaraya** built the **Ekambareswarar temple'** gopuram. The temple is uniquely marked by the presence of a 1000 pillared hallway. **Varadharaja Perumal Temple** was reconstructed by the Chola rulers. However the original evidence of the 7th century construction of the temple is not available. The temple is dedicated to Lord Vishnu and is marked by the presence of hundred pillared mandapam. The above temples along with others were added to UNESCO's tentative list of World Heritage Sites.

Q.89) Ans: C

Exp:

- The Finance Ministry recently granted 'Infrastructure' status for **exhibition and convention centres** which have been included in the latest **"Harmonised Master List of Infrastructure sub-sectors"**
- The list is notified by the **Department of Economic Affairs, Ministry of Finance** and includes various industries from the Transport, Energy, Water and Sanitation, Logistics, Communication etc. sectors.

- Inclusion in the list implies access to concessional funds, promotion of projects and continuity of construction for the specified sub-sectors. The infrastructure tag however does not include vital tax breaks.

biodiverse ecosystems and the survival of other species, including humans.

Q.90) Ans: D

Exp:

- As on date, there are 40 Sites from India in the UNESCO World heritage list. The 39th WHS is Ramappa temple, located in Palampet in Telangana's Warangal.
- At the 44th session of the UNESCO World Heritage Committee on at Fuzhou, China, Dholavira, the Harappan-era archaeological site located in Kutch district of Gujarat, was inscribed on the UNESCO list of world heritage sites, making it the first site of Indus Valley Civilisation in India to be included on the coveted list. It is the recent 40th WHS site.
- This year 6 sites were added to **India's tentative list of UNESCO world heritage sites.**
 - Hirebenakal Megalithic Site in Karnataka
 - Satpura Tiger Reserve in Madhya Pradesh
 - Temples of Kanchipuram
 - Ganga ghats in Varanasi
 - Bhedaghat-Lameta Ghat of Narmada Valley
 - Maratha military architecture in Maharashtra

Q.91) Ans: D

Exp:

About New Big 5 Project:

- It is an international initiative of **more than 250** of the world's **wildlife photographers**, conservationists and wildlife charities.
- It has been created by **British photographer Graeme Green**.
- Its aim is to **raise awareness** about the crisis facing the world's wildlife from threats including habitat loss, human-wildlife conflict, poaching, illegal wildlife trade and climate change.
- The Project has named the five animals for framing instead of firing at. These are the **elephant, polar bear, gorilla, tiger and lion**.
- All the New Big 5 animals are **keystone species**, essential to the balance of nature in their habitats,

Q.92) Ans: C

Exp:

About Christchurch Call to Action:

- The call is named after the **New Zealand city, Christchurch** where 51 members of the Muslim community were murdered in a **live-streamed terrorist attack** in 2019.
- It was launched during the **Online Extremism Summit** in Paris, France in 2019, co-chaired by New Zealand and France.



- The Christchurch Call is a **global pledge by member governments and technology partners**. Under this, they work together to **address terrorist and violent extremist contents online**.
- The pledge is **non-binding**. It consists of three sections or commitments. The governments, Online service providers and Ways in which the earlier two can work together.
- **India** is one of the **member** countries.

Q.93) Ans: B

Exp:

- **Pair 1 is correct: Helmand** is also known as **Hillmand or Helman** and, in ancient times, as **Hermand** and **Hethumand**. It is one of the **34 provinces of Afghanistan**, in the **south** of the country. It is the largest province by area. The province contains 13 districts.



- add **dialogues or chants** taken from **Bhaona** (the local traditional theatre of Assam).

Extra Edge by OnlyIAS

The puppetry is performed in three areas with distinct characteristics. These areas are:

- **The First area – Lower Assam:** Here this puppetry form is known as **Putala-Bhoariya** and influenced by the popular folk form **Bhoariya**.
- **The second area – Majuli island (Upper Assam):** Here **small wooden puppets** are used in the style of the **Ankia Naat**. (Anika Naat is an act developed by **Sankaradeva**).
- **The third area is North Assam:** Here the puppeteers use **Bangla, Assamese and Karbi songs** in their performances. These are largely borrowed from the **mobile theatre of Assam (Bhraymaman)**.

Q.95) Ans: C

Exp:

- **Statement 1 is correct:** A currency swap is a transaction in which two parties exchange an equivalent amount of money with each other but in different currencies. It is an off-balance sheet transaction. It is also known as a Cross currency swap. Often the agreement consists of swapping principal and interest payments on a loan made in one currency for principal and interest payments of a loan of equal value in another currency.
- **Statement 2 is correct:** The purpose of the Currency Risk Facility is manifold. This facility provides the flexibility to maintain an appropriate level of balance of payments or short-term liquidity. Swaps agreements between governments have supplementary objectives like the promotion of bilateral trade, maintaining the value of foreign exchange reserves with the central bank and ensuring financial stability. They are also used to obtain foreign currency loans at a better interest rate than could be obtained by borrowing directly in a foreign market. However, Countries are exposed to credit risk as either one or both the parties could default on interest and principal payments.

- **Pair 2 is incorrect:** Vorukh is an **exclave of Tajik territory**, surrounded by Kyrgyzstan. Recently clashes have broken out between Tajikistan and Kyrgyzstan with the affairs of Vorukh.
- **Pair 3 is correct:** Cali is the second-largest city in **Colombia by area** and the third most populous. It is the only major Colombian city with access to the **Pacific Coast**. Recently, the Colombian Army has tightened its control over Cali following the latest anti-government protests.

Q.94) Ans: B

Exp:

Statement 1 is incorrect: Putola Nach is **string puppetry** (not shadow puppetry) of **Assam**. 'Putul' is a word that describes both a **doll** and a **puppet** in Bengali while the word '**Naach**' means **Dance**.

Statement 2 is correct: The puppetry includes the **episodes of the Ramayana**, as well as scenes from the **Mahabharata**. Further, the puppeteers also

Q.96) Ans: A

Exp:

- **Statement 1 is correct:** Velavadar Blackbuck National Park, Bhavnagar, Gujarat is a 34-sq-km park

famous for Blackbuck. Flat land and a grassland ecosystem has helped in the protection of Blackbuck in this area. They move around in gendered groups – the males together, and the females and fawns in their own groups.

- **Statement 2 is correct:** Bishnoi is a community found in the Western Thar Desert and northern states of India. They are known for their love for nature worship and wildlife conservation. The Bishnois consider the blackbuck to be the reincarnation of their religious Guru Bhagwan Jambheshwar also known as Jambaji. Thus, they worship them.
- **Statement 3 is incorrect:** Blackbuck is now a rare species, surviving in small scattered herds in about 13 states of India. It is found from Punjab and Haryana in the north to Tamil Nadu in the south, and Rajasthan-Gujarat in the west to Odisha in the east, but nowhere in vast herds. It is notably found in the Velavadar Blackbuck National Park, Bhavnagar, Gujarat. The Tal Chhapar Blackbuck Sanctuary, Churu district, Rajasthan and Point Calimere Sanctuary in Tamil Nadu is another place to find Blackbuck. It is also found on the Deccan Plateau like The sanctuary is the Rehekuri Blackbuck Sanctuary in Ahmednagar, Maharashtra is home to the Indian Black Buck.

Q.97) Ans: C

Exp:

- **Statement 1 is Correct:** The Notified disasters covered under SDRF are Cyclone, drought, earthquake, fire, flood, tsunami, hailstorm, landslide, avalanche, cloudburst, pest attack, frost and cold waves. It is used to meet the expenditures for providing immediate relief to the victims. It was formed in accordance with the recommendations of the 13th Finance Commission.
- **Statement 2 is Incorrect:** The Centre contributes 75% of the SDRF allocation for general category States and Union Territories and 90% for special category States and Union Territories (northeastern States, Sikkim, Uttarakhand, Himachal Pradesh, Jammu and Kashmir).
- **Statement 3 is Correct:** SDRF has been constituted under the Disaster Management Act, 2005.

It is the primary fund available with the State governments for responses to notified disasters to meet expenditure for providing immediate relief. The State Disaster Relief Fund is to be audited by the CAG of India every year.

Q.98) Ans: B

Exp:

- **Statement 1 is incorrect:** Both India and Oman are members of the Indian Ocean Rim Association. Whereas, Gulf Cooperation Council (GCC) is a political and economic alliance of six Middle Eastern countries—Saudi Arabia, Kuwait, the United Arab Emirates, Qatar, Bahrain, and Oman. India is not a member of GCC.
- **Statement 2 is correct:** Al Nagah is a 14-day exercise between both the armies exchanging expertise and experience in tactics, weapon handling and firing. The exercise is aimed to enhance interoperability in counterterrorist operations in semi-urban mountainous terrain. Other exercises between India and Oman are an Air Force exercise called Eastern Bridge and a Naval Exercise called Naseem Al Bahr.
- **Statement 3 is correct:** The Port of Duqm is strategically located situated on the southeastern seaboard of Oman, overlooking Arabian Sea and Indian Ocean providing access to East Africa and also to the red sea via the Gulf of Aden. It is a deep-sea port (18 m) with a dry dock facility and comes under the Special Economic Zone at Duqm (SEZD). Oman has allowed India, including its navy, access to its Duqm port, about 550 km south of the capital Muscat. The Port of Duqm SEZ is earmarked to be the Indian Ocean's largest deep-sea port.

Q.99) Ans: B

Exp:

- **Statement 1 is correct:** The Mars Orbiter Mission/ Mangalyaan is a space probe orbiting Mars by the Indian Space Research Organisation. It is India's first interplanetary mission and it made it the fourth space agency to achieve Mars orbit, after Roscosmos, NASA, and the European Space Agency. It made India the first Asian nation to reach Martian orbit and the first nation in the world to do so on its maiden attempt.
- **Statement 2 is incorrect:** Tianwen-1 is an interplanetary mission by the China National Space Administration to send a robotic spacecraft to

Mars. Recently, it landed on Mars carrying to conduct scientific investigations into the planet's soil, geological structure, environment, atmosphere and water. With this achievement, China became the only second country after the US to send a rover to the surface of the Red Planet. NASA's Viking in 1976 made history as the first probe to land safely on Mars.

- **Statement 3 is correct:** OSIRIS-REx is NASA's first mission to visit a near-Earth asteroid, survey its surface and collect a sample from it. The spacecraft departed from asteroid Bennu, and started its two-year long journey back to Earth. As per NASA, this mission is expected to bring the largest amount of extraterrestrial material in the form of surface and dust particles back to the Earth, since the Apollo Era.

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- The European Space Agency tried to land on Mars in 2003, when its Beagle 2 probe crashed. Its ExoMars Schiaparelli spacecraft crashed in 2016 after software incorrectly estimated its altitude during an attempted landing.
- The Soviet Union made several attempts in the 1970s and its Mars 3 probe reached the surface in 1971 but only transmitted back to Earth for less than a minute before going silent.

Q.100) Ans: C

Exp:

- **Both the statements are incorrect**
- Recently, the International Energy Agency's (IEA) released its Net Zero Emissions (NZE) Roadmap - named '**Net Zero by 2050**'.
 - It is the World's first comprehensive energy roadmap which comes ahead of the United Nations Conference of the Parties (COP) 26 climate change convention in Glasgow, Scotland in November 2021.
 - **Net zero emissions'** refers to achieving an overall balance between greenhouse gas emissions produced and greenhouse gas emissions taken out of the atmosphere
- **Race to Zero** is the UN-backed global campaign rallying non-state actors – including companies, cities, regions, financial and educational institutions – to take rigorous and immediate

action to halve global emissions by 2030 and deliver a healthier, fairer, zero-carbon world in time. Race To Zero mobilizes actors outside of national governments to join the Climate Ambition Alliance.